Package 'GMPEhaz'

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Type Package

Title GMPE Functions using Norman Abrahamson's HAZ Fortran Code

Version 0.1.13.59

Depends R (>= 4.1.0)

Imports lattice

SystemRequirements GNU make

Description Provides a set of R wrapper functions to call GMPE (Ground Motion Prediction Equation) functions from Norman Abrahamson's HAZ Fortran code.

License GPL-3

LazyData TRUE

RoxygenNote 7.3.2

Suggests testthat, knitr, rmarkdown, pkgdown

Config/testthat/edition 3

URL https://personlin.github.io/GMPEhaz/, https://github.com/personlin/GMPEhaz

BugReports https://github.com/personlin/GMPEhaz/issues

VignetteBuilder knitr

Encoding UTF-8

NeedsCompilation yes

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`AB03` *GMPE function for Atkinson and Boore(2003)*

Description

`AB03` returns the ground-motion prediction with it sigma of Atkinson and Boore(2003) GMPE.

Usage

```
AB03(Mag, Rrup, Prd, ftype = 0, depth, Sc = 0, Sd = 0, Se = 0)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`ftype`	style of faulting.
`depth`	hypocentral depth(km).

Sc	NEHRP B Sc=0, Sd=0, Se=0, Vs>760m/s, NEHRP C Sc=1, Sd=0, Se=0, 360<Vs<760
Sd	NEHRP D Sc=0, Sd=1, Se=0, 180<Vs<360
Se	NEHRP E Sc=0, Sd=0, Se=1, Vs<180

Details

Atkinson, G., and D. Boore (2003), Empirical ground-motion relations for subduction-zone earthquakes and their application to cascadia and other regions, Bulletin of the Seismological Society of America, 93(4), 1703-1729. doi:10.1785/0120020156

Value

A list will be return, including mag, Rrup, lnY, sigma, specT, period, iflag, ftype, depth, Sc, Sd, Se.

Examples

```
AB03(6, 20, 0, 0, 10, 0, 0, 0)
AB03(7, 10, 0, 0, 10, 0, 0, 0)
```

AC10 *GMPE function for Akkar and Cagnan(2010)*

Description

`AC10` returns the ground-motion prediction with it sigma of Akkar and Cagnan(2010) GMPE.

Usage

```
AC10(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Akkar, S., and Z. Cagnan (2010), A local ground-motion predictive model for turkey, and its comparison with other regional and global ground-motion models, Bulletin of the Seismological Society of America, 100(6), 2978-2995. doi:10.1785/0120090367

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, Vs30, ftype, pga4nl.

Examples

```
AC10(6, 20, 0, 0, 760)
AC10(7, 10, 0, 0, 760)
```

AGA16.tw.C01 *GMPE function for adjusted Abrahamson et al.(2016)*

Description

AGA16.tw.C01 returns the ground-motion prediction with it sigma of Abrahamson et al.(2016) GMPE adjusted to Taiwan.

Usage

```
AGA16.tw.C01(Mag, Rrup, Prd, ftype = 0, Vs30, Ztor, depth, Rhypo)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface (use rupture distance), ftype=1 for intraslab (use hypocentral distance)
Vs30	Vs30(m/s).
Ztor	Depth to the top of the finite rupture model (km).
depth	hypocentral depth(km).
Rhypo	hypocentral distance(km)

Details

Norman Abrahamson, Nicholas Gregor, and Kofi Addo(2016) BC Hydro Ground Motion Prediction Equations for Subduction Earthquakes, Earthquake Spectra, Vol. 32, No. 1, pp. 23-44. doi:10.1193/051712EQS188MR

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo, deltac1.

Examples

```
AGA16.tw.C01(6, 20, 0, 0, 760, 3, 10, 20)
AGA16.tw.C01(7, 10, 0, 0, 760, 0, 10, 20)
```

AGA16.tw.F10 *GMPE function for adjusted Abrahamson et al.(2016)*

Description

AGA16.tw.F10 returns the ground-motion prediction with it sigma of Abrahamson et al.(2016) GMPE adjusted to Taiwan.

Usage

```
AGA16.tw.F10(Mag, Rrup, Prd, ftype = 0, Vs30, forearc = 0, Ztor, depth, Rhypo)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface (use rupture distance), ftype=1 for intraslab (use hypocentral distance)
Vs30	Vs30(m/s).
forearc	0 = Forearc site, 1 = Backarc site
Ztor	Depth to the top of the finite rupture model (km).
depth	hypocentral depth(km).
Rhypo	hypocentral distance(km)

Details

Norman Abrahamson, Nicholas Gregor, and Kofi Addo(2016) BC Hydro Ground Motion Prediction Equations for Subduction Earthquakes, Earthquake Spectra, Vol. 32, No. 1, pp. 23-44. doi:10.1193/051712EQS188MR

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo, deltac1.

Examples

```
AGA16.tw.F10(6, 20, 0, 0, 760, 0, 3, 10, 20)
AGA16.tw.F10(7, 10, 0, 0, 760, 0, 0, 10, 20)
```

AM09 *GMPE function for Atkinson and Macias (2009)*

Description

AM09 returns the ground-motion prediction with it sigma of Atkinson and Macias (2009) GMPE. (Cascadia Subduction interface, Horizontal NEHRP B/C)

Usage

```
AM09(Mag, Rrup, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.

Details

Atkinson, G. M., and Macias, M. (2009). Predicted Ground Motions for Great Interface Earthquakes in the Cascadia Subduction Zone. Bulletin of the Seismological Society of America, 99(3), 1552-1578. doi:10.1785/0120080147

Value

A list will be return, including mag, Rrup, lnY, sigma, specT, period, iflag.

Examples

```
AM09(6, 20, 0)
AM09(7, 10, 0)
```

Arroyo2010 *GMPE function for Arroyo et al. (2010)*

Description

Arroyo2010 returns the ground-motion prediction with it sigma of Arroyo et al. (2010) GMPE.

Usage

```
Arroyo2010(Mag, Rrup, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.

Details

Arroyo, D. Garcia, M. Ordaz, M. A. Mora, and S. K. Singh. Strong ground-motion relations for Mexican interplate earthquakes. Journal of Seismology, 14(4):769-785, Oct 2010. https://link.springer.com/article/10.1007/s10950-010-9200-0

Value

A list will be return, including mag, Rrup, lnY, specT, iflag, period, phi1, tau1, sigma.

Examples

```
Arroyo2010(6, 20, 0)
Arroyo2010(7, 10, 0)
```

ASB14 *GMPE function for Akkar et al.(2013)*

Description

ASB14 returns the ground-motion prediction with it sigma of Akkar et al.(2014) GMPE.

Usage

```
ASB14(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Akkar, S., M. a. Sandikkaya, and J. J. Bommer(2014), Empirical ground-motion models for point- and extended-source crustal earthquake scenarios in europe and the middle east, Bulletin of Earthquake Engineering, Volume 12, Issue 1, pp 359-387. doi:10.1007/s1051801394614

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, ftype, Vs30, phi, tau.

Examples

```
ASB14(6, 20, 0, 0, 760)
ASB14(7, 10, 0, 0, 760)
```

ASB14.tw.B01 *GMPE function for Akkar et al.(2013)*

Description

ASB14.tw.B01 returns the ground-motion prediction with it sigma of Akkar et al.(2014) GMPE adjusted to Taiwan.

Usage

```
ASB14.tw.B01(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Akkar, S., M. a. Sandikkaya, and J. J. Bommer(2014), Empirical ground-motion models for point- and extended-source crustal earthquake scenarios in europe and the middle east, Bulletin of Earthquake Engineering, Volume 12, Issue 1, pp 359-387. doi:10.1007/s1051801394614

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, ftype, Vs30, phi, tau.

Examples

```
ASB14.tw.B01(6, 20, 0, 0, 760)
ASB14.tw.B01(7, 10, 0, 0, 760)
```

ASB14.tw.C01 *GMPE function for Akkar et al.(2013)*

Description

ASB14.tw.C01 returns the ground-motion prediction with it sigma of Akkar et al.(2014) GMPE adjusted to Taiwan.

Usage

```
ASB14.tw.C01(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Akkar, S., M. a. Sandikkaya, and J. J. Bommer(2014), Empirical ground-motion models for point- and extended-source crustal earthquake scenarios in europe and the middle east, Bulletin of Earthquake Engineering, Volume 12, Issue 1, pp 359-387. doi:10.1007/s1051801394614

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, ftype, Vs30, phi, tau.

Examples

```
ASB14.tw.C01(6, 20, 0, 0, 760)
ASB14.tw.C01(7, 10, 0, 0, 760)
```

ASB14.tw.E02 *GMPE function for Akkar et al.(2013)*

Description

ASB14.tw.E02 returns the ground-motion prediction with it sigma of Akkar et al.(2014) GMPE adjusted to Taiwan.

Usage

```
ASB14.tw.E02(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Akkar, S., M. a. Sandikkaya, and J. J. Bommer(2014), Empirical ground-motion models for point- and extended-source crustal earthquake scenarios in europe and the middle east, Bulletin of Earthquake Engineering, Volume 12, Issue 1, pp 359-387. doi:10.1007/s1051801394614

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, ftype, Vs30, phi, tau.

Examples

```
ASB14.tw.E02(6, 20, 0, 0, 760)
ASB14.tw.E02(7, 10, 0, 0, 760)
```

ASK14 *GMPE function for Abrahamson et al.(2014) NGA-West2 model*

Description

ASK14 returns the ground-motion prediction with it sigma of Abrahamson et al.(2014) GMPE.

Usage

```
ASK14(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Dip,
  ftype = 0,
  rupwidth,
  Vs30 = 760,
  hwflag = 0,
  Ztor,
  Vs30_class = 1,
  Z1.0,
  Rx,
  Ry0,
  regionflag = 0,
  msasflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Dip	Dip angle of the fault plane.
ftype	style of faulting.
rupwidth	Down-dip rupture width (km).
Vs30	Vs30(m/s).
hwflag	hanging-wall flag, 1 for hanging-wall.

`Ztor`	Depth to the top of the finite rupture model (km).
`Vs30_class`	1 measured, 0 estimated.
`Z1.0`	the depth to the shear-wave velocity horizon of 1.0(km/s).
`Rx`	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
`Ry0`	Horizontal distance(km) off the end of the rupture measured parallel to strike.
`regionflag`	Regional attenuation flag, 0 = Global, 1 = Taiwan, 2 = China, 3 = Japan.
`msasflag`	Mainshock and Aftershocks flag, 0 = Mainshocks, 1 = Aftershocks.

Details

Abrahamson, N. A., W. J. Silva, and R. Kamai (2014), Summary of the ASK14 ground motion relation for active crustal regions, Earthquake Spectra, 30(3), 1025-1055. doi:10.1193/070913EQS198M

Value

A list will be return, including mag, dip, ftype, rupwidth, Rrup, Rjb, Vs30, hwflag, lnY, sigma, specT, period, Ztor, iflag, vs30_class, z10, Rx, Ry0, regionflag, msasflag, phi, tau.

Examples

```
ASK14(6, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
ASK14(7, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
```

`ASK14.tw.B01` *GMPE function for Abrahamson et al.(2014) NGA-West2 model*

Description

`ASK14.tw.B01` returns the ground-motion prediction with it sigma of Abrahamson et al.(2014) GMPE adjusted to Taiwan.

Usage

```
ASK14.tw.B01(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Dip,
  ftype = 0,
  rupwidth,
  Vs30 = 760,
  hwflag = 0,
  Ztor,
  Vs30_class = 1,
  Z1.0,
  Rx,
  Ry0,
```

```
  regionflag = 0,
  msasflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Dip	Dip angle of the fault plane.
ftype	style of faulting.
rupwidth	Down-dip rupture width (km).
Vs30	Vs30(m/s).
hwflag	hanging-wall flag, 1 for hanging-wall.
Ztor	Depth to the top of the finite rupture model (km).
Vs30_class	1 measured, 0 estimated.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
Ry0	Horizontal distance(km) off the end of the rupture measured parallel to strike.
regionflag	Regional attenuation flag, 0 = Global, 1 = Taiwan, 2 = China, 3 = Japan.
msasflag	Mainshock and Aftershocks flag, 0 = Mainshocks, 1 = Aftershocks.

Details

Abrahamson, N. A., W. J. Silva, and R. Kamai (2014), Summary of the ASK14 ground motion relation for active crustal regions, Earthquake Spectra, 30(3), 1025-1055. doi:10.1193/070913EQS198M

Value

A list will be return, including mag, dip, ftype, rupwidth, Rrup, Rjb, Vs30, hwflag, lnY, sigma, specT, period, Ztor, iflag, vs30_class, z10, Rx, Ry0, regionflag, msasflag, phi, tau.

Examples

```
ASK14.tw.B01(6, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
ASK14.tw.B01(7, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
```

ASK14.tw.C01 *GMPE function for Abrahamson et al.(2014) NGA-West2 model*

Description

ASK14.tw.C01 returns the ground-motion prediction with it sigma of Abrahamson et al.(2014) GMPE adjusted to Taiwan.

Usage

```
ASK14.tw.C01(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Dip,
  ftype = 0,
  rupwidth,
  Vs30 = 760,
  hwflag = 0,
  Ztor,
  Vs30_class = 1,
  Z1.0,
  Rx,
  Ry0,
  regionflag = 0,
  msasflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Dip	Dip angle of the fault plane.
ftype	style of faulting.
rupwidth	Down-dip rupture width (km).
Vs30	Vs30(m/s).
hwflag	hanging-wall flag, 1 for hanging-wall.
Ztor	Depth to the top of the finite rupture model (km).
Vs30_class	1 measured, 0 estimated.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
Ry0	Horizontal distance(km) off the end of the rupture measured parallel to strike.
regionflag	Regional attenuation flag, 0 = Global, 1 = Taiwan, 2 = China, 3 = Japan.
msasflag	Mainshock and Aftershocks flag, 0 = Mainshocks, 1 = Aftershocks.

Details

Abrahamson, N. A., W. J. Silva, and R. Kamai (2014), Summary of the ASK14 ground motion relation for active crustal regions, Earthquake Spectra, 30(3), 1025-1055. doi:10.1193/070913EQS198M

Value

A list will be return, including mag, dip, ftype, rupwidth, Rrup, Rjb, Vs30, hwflag, lnY, sigma, specT, period, Ztor, iflag, vs30_class, z10, Rx, Ry0, regionflag, msasflag, phi, tau.

Examples

```
ASK14.tw.C01(6, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
ASK14.tw.C01(7, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
```

ASK14.tw.E03 *GMPE function for Abrahamson et al.(2014) NGA-West2 model*

Description

`ASK14.tw.E03` returns the ground-motion prediction with it sigma of Abrahamson et al.(2014) GMPE adjusted to Taiwan.

Usage

```
ASK14.tw.E03(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Dip,
  ftype = 0,
  rupwidth,
  Vs30 = 760,
  hwflag = 0,
  Ztor,
  Vs30_class = 1,
  Z1.0,
  Rx,
  Ry0,
  regionflag = 0,
  msasflag = 0
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Prd`	Period of spectral acceleration.

Dip	Dip angle of the fault plane.
ftype	style of faulting.
rupwidth	Down-dip rupture width (km).
Vs30	Vs30(m/s).
hwflag	hanging-wall flag, 1 for hanging-wall.
Ztor	Depth to the top of the finite rupture model (km).
Vs30_class	1 measured, 0 estimated.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
Ry0	Horizontal distance(km) off the end of the rupture measured parallel to strike.
regionflag	Regional attenuation flag, 0 = Global, 1 = Taiwan, 2 = China, 3 = Japan.
msasflag	Mainshock and Aftershocks flag, 0 = Mainshocks, 1 = Aftershocks.

Details

Abrahamson, N. A., W. J. Silva, and R. Kamai (2014), Summary of the ASK14 ground motion relation for active crustal regions, Earthquake Spectra, 30(3), 1025-1055. doi:10.1193/070913EQS198M

Value

A list will be return, including mag, dip, ftype, rupwidth, Rrup, Rjb, Vs30, hwflag, lnY, sigma, specT, period, Ztor, iflag, vs30_class, z10, Rx, Ry0, regionflag, msasflag, phi, tau.

Examples

```
ASK14.tw.E03(6, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
ASK14.tw.E03(7, 20, 20, 0, 90, 0, 10, 760, 0, 0, 1, 0.5, 20, 10, 0, 0)
```

---

atten2 *Wrap function for all GMPE*

Description

atten2 returns the ground-motion prediction with it sigma of any GMPE.

Usage

```
atten2(
  GMPEname,
  Periods,
  Mag,
  Rrup,
  Ztor = 0,
  RV = 0,
  depth,
  ftype = 0,
  plot = FALSE
)
```

Arguments

GMPEname	Name of GMPE.
Periods	Periods of response spetra
Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
RV	Indicator of reverse fault.
depth	Focal depth.
ftype	style of fault.
plot	flag for plot or not.

Value

A list will be return, including mag, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, phi, tau.

Examples

```
atten2("LL08Rock", 0, Mag=c(6,7,8), 20, 0, 0, 10, 0, FALSE)
atten2("LL08Rock", Periods=c(0.01, 0.1, 0.2, 0.5, 1, 2, 5), 6, 10, 0, 0, 10, 0, FALSE)
```

BCHydroSub2018Global *GMPE function for Abrahamson et al.(2018)*

Description

BCHydroSub2018Global returns the ground-motion prediction with it sigma of Abrahamson et al.(2016) GMPE.

Usage

```
BCHydroSub2018Global(
  Mag,
  Rrup,
  Prd,
  ftype = 0,
  Vs30,
  forearc = 1,
  depth,
  Rhypo
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`ftype`	fytpe=0 for interface (use rupture distance), ftype=1 for intraslab (use hypocentral distance)
`Vs30`	Vs30(m/s).
`forearc`	0 = Forearc site, 1 = Backarc site
`depth`	hypocentral depth(km).
`Rhypo`	hypocentral distance(km)

Details

Norman Abrahamson, Nicholas Gregor, and Kofi Addo(2016) BC Hydro Ground Motion Prediction Equations for Subduction Earthquakes, Earthquake Spectra, Vol. 32, No. 1, pp. 23-44. doi:10.1193/051712EQS188MR

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo, deltac1.

Examples

```
BCHydroSub2018Global(6, 20, 0, 0, 760, 1, 10, 20)
BCHydroSub2018Global(7, 10, 0, 0, 760, 1, 10, 20)
```

`BCHydroSub2018SouthAM` *GMPE function for Abrahamson et al.(2018)*

Description

`BCHydroSub2018SouthAM` returns the ground-motion prediction with it sigma of Abrahamson et al.(2016) GMPE.

Usage

```
BCHydroSub2018SouthAM(
  Mag,
  Rrup,
  Prd,
  ftype = 0,
  Vs30,
  forearc = 1,
  depth,
  Rhypo
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface (use rupture distance), ftype=1 for intraslab (use hypocentral distance)
Vs30	Vs30(m/s).
forearc	0 = Forearc site, 1 = Backarc site
depth	hypocentral depth(km).
Rhypo	hypocentral distance(km)

Details

Norman Abrahamson, Nicholas Gregor, and Kofi Addo(2016) BC Hydro Ground Motion Prediction Equations for Subduction Earthquakes, Earthquake Spectra, Vol. 32, No. 1, pp. 23-44. doi:10.1193/051712EQS188MR

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo, deltac1.

Examples

```
BCHydroSub2018SouthAM(6, 20, 0, 0, 760, 1, 10, 20)
BCHydroSub2018SouthAM(7, 10, 0, 0, 760, 1, 10, 20)
```

BCHydroSub2018TW *GMPE function for Abrahamson et al.(2018)*

Description

BCHydroSub2018TW returns the ground-motion prediction with it sigma of Abrahamson et al.(2016) GMPE.

Usage

```
BCHydroSub2018TW(Mag, Rrup, Prd, ftype = 0, Vs30, forearc = 1, depth, Rhypo)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface (use rupture distance), ftype=1 for intraslab (use hypocentral distance)
Vs30	Vs30(m/s).
forearc	0 = Forearc site, 1 = Backarc site
depth	hypocentral depth(km).
Rhypo	hypocentral distance(km)

Details

Norman Abrahamson, Nicholas Gregor, and Kofi Addo(2016) BC Hydro Ground Motion Prediction Equations for Subduction Earthquakes, Earthquake Spectra, Vol. 32, No. 1, pp. 23-44. doi:10.1193/051712EQS188MR

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo, deltac1.

Examples

```
BCHydroSub2018TW(6, 20, 0, 0, 760, 1, 10, 20)
BCHydroSub2018TW(7, 10, 0, 0, 760, 1, 10, 20)
```

BCHydroSubV3 *GMPE function for Abrahamson et al.(2016)*

Description

`BCHydroSubV3` returns the ground-motion prediction with it sigma of Abrahamson et al.(2016) GMPE.

Usage

```
BCHydroSubV3(Mag, Rrup, Prd, ftype = 0, Vs30, forearc = 1, depth, Rhypo)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`ftype`	fytpe=0 for interface (use rupture distance), ftype=1 for intraslab (use hypocentral distance)
`Vs30`	Vs30(m/s).
`forearc`	0 = Forearc site, 1 = Backarc site
`depth`	hypocentral depth(km).
`Rhypo`	hypocentral distance(km)

Details

Norman Abrahamson, Nicholas Gregor, and Kofi Addo(2016) BC Hydro Ground Motion Prediction Equations for Subduction Earthquakes, Earthquake Spectra, Vol. 32, No. 1, pp. 23-44. doi:10.1193/051712EQS188MR

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo, deltac1.

Examples

```
BCHydroSubV3(6, 20, 0, 0, 760, 1, 10, 20)
BCHydroSubV3(7, 10, 0, 0, 760, 1, 10, 20)
```

BI14 *GMPE function for Bindi et al(2014)*

Description

BI14 returns the ground-motion prediction with it sigma of Bindi et al.(2014) GMPE.

Usage

```
BI14(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Bindi, D., M. Massa, L. Luzi, G. Ameri, F. Pacor, R. Puglia, and P. Augliera (2014), Pan-european ground-motion prediction equations for the average horizontal component of pGA, pGV, and 5 dataset, Bulletin of Earthquake Engineering, 12(1), 391-430. doi:10.1007/s1051801395255

Value

A list will be return, including mag, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, phi, tau.

Examples

```
BI14(6, 20, 0, 0, 760)
BI14(7, 10, 0, 0, 760)
```

`BI14.tw.B01` *GMPE function for Bindi et al(2014)*

Description

`BI14.tw.B01` returns the ground-motion prediction with it sigma of Bindi et al.(2014) GMPE adjusted to Taiwan.

Usage

```
BI14.tw.B01(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`ftype`	style of faulting.
`Vs30`	Vs30(m/s).

Details

Bindi, D., M. Massa, L. Luzi, G. Ameri, F. Pacor, R. Puglia, and P. Augliera (2014), Pan-european ground-motion prediction equations for the average horizontal component of pGA, pGV, and 5 dataset, Bulletin of Earthquake Engineering, 12(1), 391-430. doi:10.1007/s1051801395255

Value

A list will be return, including mag, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, phi, tau.

Examples

```
BI14.tw.B01(6, 20, 0, 0, 760)
BI14.tw.B01(7, 10, 0, 0, 760)
```

`BI14.tw.C01` *GMPE function for Bindi et al(2014)*

Description

`BI14.tw.C01` returns the ground-motion prediction with it sigma of Bindi et al.(2014) GMPE adjusted to Taiwan.

Usage

```
BI14.tw.C01(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Bindi, D., M. Massa, L. Luzi, G. Ameri, F. Pacor, R. Puglia, and P. Augliera (2014), Pan-european ground-motion prediction equations for the average horizontal component of pGA, pGV, and 5 dataset, Bulletin of Earthquake Engineering, 12(1), 391-430. doi:10.1007/s1051801395255

Value

A list will be return, including mag, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, phi, tau.

Examples

```
BI14.tw.C01(6, 20, 0, 0, 760)
BI14.tw.C01(7, 10, 0, 0, 760)
```

BI14.tw.E02 *GMPE function for Bindi et al(2014)*

Description

BI14.tw.E02 returns the ground-motion prediction with it sigma of Bindi et al.(2014) GMPE adjusted to Taiwan.

Usage

```
BI14.tw.E02(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Bindi, D., M. Massa, L. Luzi, G. Ameri, F. Pacor, R. Puglia, and P. Augliera (2014), Pan-european ground-motion prediction equations for the average horizontal component of pGA, pGV, and 5 dataset, Bulletin of Earthquake Engineering, 12(1), 391-430. doi:10.1007/s1051801395255

Value

A list will be return, including mag, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, phi, tau.

Examples

```
BI14.tw.E02(6, 20, 0, 0, 760)
BI14.tw.E02(7, 10, 0, 0, 760)
```

BSSA14 *GMPE function for Boore et al.(2014) NGA-West2 model*

Description

BSSA14 returns the ground-motion prediction with it sigma of Boore et al.(2014) GMPE.

Usage

```
BSSA14(Mag, Rjb, Prd, Vs30, ftype = 0, Z1.0, regionflag = 0, basinflag = 0)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
regionflag	Regional attenuation flag, 0 = Global, 1 = China-Turkey, 2 = Italy-Japan.
basinflag	Basin Adjustments.

Details

Boore, D. M., J. P. Stewart, E. Seyhan, and G. M. Atkinson (2014), NGA-west2 equations for predicting PGA, PGV, and 5 doi:10.1193/070113EQS184M

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, Vs30, ftype, pga4nl, Z10, regionflag, basinflag, phi, tau.

Examples

```
BSSA14(6, 20, 0, 760, 0, 0.5, 0, 0)
BSSA14(7, 20, 0, 760, 0, 0.5, 0, 0)
```

BSSA14.tw.B01 *GMPE function for Boore et al.(2014) NGA-West2 model*

Description

BSSA14.tw.B01 returns the ground-motion prediction with it sigma of Boore et al.(2014) GMPE adjusted to Taiwan.

Usage

```
BSSA14.tw.B01(
  Mag,
  Rjb,
  Prd,
  Vs30,
  ftype = 0,
  Z1.0,
  regionflag = 0,
  basinflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
regionflag	Regional attenuation flag, 0 = Global, 1 = China-Turkey, 2 = Italy-Japan.
basinflag	Basin Adjustments.

Details

Boore, D. M., J. P. Stewart, E. Seyhan, and G. M. Atkinson (2014), NGA-west2 equations for predicting PGA, PGV, and 5 doi:10.1193/070113EQS184M

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, Vs30, ftype, pga4nl, Z10, regionflag, basinflag, phi, tau.

Examples

```
BSSA14.tw.B01(6, 20, 0, 760, 0, 0.5, 0, 0)
BSSA14.tw.B01(7, 20, 0, 760, 0, 0.5, 0, 0)
```

BSSA14.tw.C01 *GMPE function for Boore et al.(2014) NGA-West2 model*

Description

BSSA14.tw.C01 returns the ground-motion prediction with it sigma of Boore et al.(2014) GMPE adjusted to Taiwan.

Usage

```
BSSA14.tw.C01(
  Mag,
  Rjb,
  Prd,
  Vs30,
  ftype = 0,
  Z1.0,
  regionflag = 0,
  basinflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
regionflag	Regional attenuation flag, 0 = Global, 1 = China-Turkey, 2 = Italy-Japan.
basinflag	Basin Adjustments.

Details

Boore, D. M., J. P. Stewart, E. Seyhan, and G. M. Atkinson (2014), NGA-west2 equations for predicting PGA, PGV, and 5 doi:10.1193/070113EQS184M

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, Vs30, ftype, pga4nl, Z10, regionflag, basinflag, phi, tau.

Examples

```
BSSA14.tw.C01(6, 20, 0, 760, 0, 0.5, 0, 0)
BSSA14.tw.C01(7, 20, 0, 760, 0, 0.5, 0, 0)
```

BSSA14.tw.E02 *GMPE function for Boore et al.(2014) NGA-West2 model*

Description

BSSA14.tw.E02 returns the ground-motion prediction with it sigma of Boore et al.(2014) GMPE adjusted to Taiwan.

Usage

```
BSSA14.tw.E02(
  Mag,
  Rjb,
  Prd,
  Vs30,
  ftype = 0,
  Z1.0,
  regionflag = 0,
  basinflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
regionflag	Regional attenuation flag, 0 = Global, 1 = China-Turkey, 2 = Italy-Japan.
basinflag	Basin Adjustments.

Details

Boore, D. M., J. P. Stewart, E. Seyhan, and G. M. Atkinson (2014), NGA-west2 equations for predicting PGA, PGV, and 5 doi:10.1193/070113EQS184M

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, Vs30, ftype, pga4nl, Z10, regionflag, basinflag, phi, tau.

Examples

```
BSSA14.tw.E02(6, 20, 0, 760, 0, 0.5, 0, 0)
BSSA14.tw.E02(7, 20, 0, 760, 0, 0.5, 0, 0)
```

Camp03H *GMPE function for Campbell and Bozorgnia(2003)*

Description

Camp03H returns the ground-motion prediction with it sigma of Campbell and Bozorgnia(2003) GMPE.

Usage

```
Camp03H(Mag, Rjb, Rseis, Prd, Svfs, Ssr, Sfr, Frv, Fth, hwflag = 0)
```

Arguments

Mag	Earthquake local magnitude, Numeric.
Rjb	Rupture distance(km), Numeric.
Rseis	Seismogenic distance(km)
Prd	Period of spectral acceleration.
Svfs	Svfs
Ssr	Ssr
Sfr	Sfr
Frv	Frv
Fth	Fth
hwflag	hanging-wall flag, 1 for hanging-wall.

Details

Campbell, Kenneth W. and Bozorgnia, Yousef (2003), Updated Near-Source Ground-Motion (Attenuation) Relations for the Horizontal and Vertical Components of Peak Ground Acceleration and Acceleration Response Spectra, Bulletin of the Seismological Society of America, 93(1), 314-331. doi:10.1785/0120020029

Value

A list will be return, including mag, Rseis, Rjb, lnY, sigma, specT, period, Svfs, Ssr, Sfr, Frv, Fth, hwflag, iflag.

Examples

```
Camp03H(6, 20, 20, 0, 0, 0, 0, 0, 0)
Camp03H(7, 10, 20, 0, 0, 0, 0, 0, 0)
```

CB14 *GMPE function for Campbell and Bozorgnia(2014) NGA-West2 model*

Description

CB14 returns the ground-motion prediction with it sigma of Campbell and Bozorgnia(2014) GMPE.

Usage

```
CB14(
  Mag,
  Rrup,
  Rjb,
  ftype = 0,
  Prd,
  Vs30 = 760,
  Ztor,
  Z2.5,
  Dip,
  depth,
  hwflag = 0,
  Rx,
  rupwidth,
  regionflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner-Boore distance(km), Numeric.
ftype	style of faulting.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Ztor	Depth to the top of the finite rupture model (km).
Z2.5	the depth to the shear-wave velocity horizon of 2.5(km/s).
Dip	Dip angle of the fault plane.
depth	hypocentral depth(km).
hwflag	hanging-wall flag, 1 for hanging-wall.
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
rupwidth	Down-dip rupture width (km).
regionflag	Regional attenuation flag, 0 = California, 1 = Japan 2 = China, 3 = Italy.

Details

Campbell, K. W., and Y. Bozorgnia (2014), NGA-west2 ground motion model for the average horizontal components of PGA, PGV, and 5 Earthquake Spectra, 30(3), 1087-1115. doi:10.1193/062913EQS175M

Value

A list will be return, including mag, Rrup, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, Ztor, Z25, dip, depth, hwflag, Rx, rupwidth, regionflag, phi, tau.

Examples

```
CB14(6, 20, 20, 0, 0, 760, 0, 1.5, 90, 10, 0, 20, 10, 0)
CB14(7, 20, 20, 0, 0, 760, 0, 1.5, 90, 10, 0, 20, 10, 0)
```

CB14.tw.C01 *GMPE function for adjusted Campbell and Bozorgnia(2014) NGA-West2 model*

Description

`CB14.tw.C01` returns the ground-motion prediction with it sigma of Campbell and Bozorgnia(2014) GMPE adjusted to Taiwan..

Usage

```
CB14.tw.C01(
  Mag,
  Rrup,
  Rjb,
  ftype = 0,
  Prd,
  Vs30 = 760,
  Ztor,
  Z2.5,
  Dip,
  depth,
  hwflag = 0,
  Rx,
  rupwidth,
  regionflag = 0
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner-Boore distance(km), Numeric.
`ftype`	style of faulting.
`Prd`	Period of spectral acceleration.
`Vs30`	Vs30(m/s).
`Ztor`	Depth to the top of the finite rupture model (km).
`Z2.5`	the depth to the shear-wave velocity horizon of 2.5(km/s).
`Dip`	Dip angle of the fault plane.

depth	hypocentral depth(km).
hwflag	hanging-wall flag, 1 for hanging-wall.
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
rupwidth	Down-dip rupture width (km).
regionflag	Regional attenuation flag, 0 = California, 1 = Japan 2 = China, 3 = Italy.

Details

Campbell, K. W., and Y. Bozorgnia (2014), NGA-west2 ground motion model for the average horizontal components of PGA, PGV, and 5 Earthquake Spectra, 30(3), 1087-1115. doi:10.1193/062913EQS175M

Value

A list will be return, including mag, Rrup, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, Ztor, Z25, dip, depth, hwflag, Rx, rupwidth, regionflag, phi, tau.

Examples

```
CB14.tw.C01(6, 20, 20, 0, 0, 760, 0, 1.5, 90, 10, 0, 20, 10, 0)
CB14.tw.C01(7, 20, 20, 0, 0, 760, 0, 1.5, 90, 10, 0, 20, 10, 0)
```

CB14.tw.E05 *GMPE function for adjusted Campbell and Bozorgnia(2014) NGA-West2 model*

Description

CB14.tw.E05 returns the ground-motion prediction with it sigma of Campbell and Bozorgnia(2014) GMPE adjusted to Taiwan..

Usage

```
CB14.tw.E05(
  Mag,
  Rrup,
  Rjb,
  ftype = 0,
  Prd,
  Vs30 = 760,
  Ztor,
  Z2.5,
  Dip,
  depth,
  hwflag = 0,
  Rx,
  rupwidth,
  regionflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner-Boore distance(km), Numeric.
ftype	style of faulting.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Ztor	Depth to the top of the finite rupture model (km).
Z2.5	the depth to the shear-wave velocity horizon of 2.5(km/s).
Dip	Dip angle of the fault plane.
depth	hypocentral depth(km).
hwflag	hanging-wall flag, 1 for hanging-wall.
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
rupwidth	Down-dip rupture width (km).
regionflag	Regional attenuation flag, 0 = California, 1 = Japan 2 = China, 3 = Italy.

Details

Campbell, K. W., and Y. Bozorgnia (2014), NGA-west2 ground motion model for the average horizontal components of PGA, PGV, and 5 Earthquake Spectra, 30(3), 1087-1115. doi:10.1193/062913EQS175M

Value

A list will be return, including mag, Rrup, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, Ztor, Z25, dip, depth, hwflag, Rx, rupwidth, regionflag, phi, tau.

Examples

```
CB14.tw.E05(6, 20, 20, 0, 0, 760, 0, 1.5, 90, 10, 0, 20, 10, 0)
CB14.tw.E05(7, 20, 20, 0, 0, 760, 0, 1.5, 90, 10, 0, 20, 10, 0)
```

Chao2017 *GMPE function for Chao et al.(2017)*

Description

Chao2017 returns the ground-motion prediction with it sigma of Chao et al.(2017) GMPE.

Usage

```
Chao2017(
  Mag,
  Rrup,
  Prd = 0,
  ftype = 0,
  Vs30 = 760,
  Vs30_class = 0,
  Ztor = 0,
  Z1.0 = 0.0058959,
  sourcetype = 0,
  msasflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting, -1 for Normal, 0 for Strike-Slip, 1 for Reverse for sourcetype = 0. 0 for interface, 1 for intraslab for soucetype = 1.
Vs30	Vs30(m/s).
Vs30_class	Vs30 data class, 0 for estimated, 1 for measured.
Ztor	Depth to the top of the finite rupture model (km).
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
sourcetype	source type, 0 for crustal, 1 for Subduction.
msasflag	Mainshock and Aftershocks flag, 0 = Mainshocks, 1 = Aftershocks.

Details

Chao et al.(2017)

Value

A list will be return, including mag, Rrup, ftype, lnY, sigma, sclass, specT, attenName, period, iflag, sourcetype, depth, phi, tau.

Examples

```
Chao2017(6, 20, 0, 0, 760, 0, 0, 0, 0)
Chao2017(7, 10, 0, 0, 760, 0, 0, 0, 0)
```

Chao2018 *GMPE function for Chao et al.(2018)*

Description

Chao2018 returns the ground-motion prediction with it sigma of Chao et al.(2018) GMPE.

Usage

```
Chao2018(
  Mag,
  Rrup,
  Prd = 0,
  ftype = 0,
  Vs30 = 760,
  Vs30_class = 0,
  Ztor = 0,
  Z1.0 = 0.0058959,
  sourcetype = 0,
  msasflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting, -1 for Normal, 0 for Strike-Slip, 1 for Reverse for sourcetype = 0. 0 for interface, 1 for intraslab for soucetype = 1.
Vs30	Vs30(m/s).
Vs30_class	Vs30 data class, 0 for estimated, 1 for measured.
Ztor	Depth to the top of the finite rupture model (km).
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
sourcetype	source type, 0 for crustal, 1 for Subduction.
msasflag	Mainshock and Aftershocks flag, 0 = Mainshocks, 1 = Aftershocks.

Details

Chao et al.(2018)

Value

A list will be return, including mag, Rrup, ftype, lnY, sigma, sclass, specT, attenName, period, iflag, sourcetype, depth, phi, tau.

Examples

```
Chao2018(6, 20, 0, 0, 760, 0, 0, 0, 0)
Chao2018(7, 10, 0, 0, 760, 0, 0, 0, 0)
```

cnames.AB03 *Variable name of AB03 GMPEs.*

Description

Variable name of AB03 GMPEs.

Usage

```
cnames.AB03
```

Format

Vectors of AB03 GMPE.

cnames.AC10 *Variable name of AC10 GMPEs.*

Description

Variable name of AC10 GMPEs.

Usage

```
cnames.AC10
```

Format

Vectors of AC10 GMPE.

cnames.AGA16.tw.C01 *Variable name of AGA16.tw.C01 GMPEs.*

Description

Variable name of AGA16.tw.C01 GMPEs.

Usage

```
cnames.AGA16.tw.C01
```

Format

Vectors of AGA16.tw.C01 GMPE.

cnames.AGA16.tw.F10 *Variable name of AGA16.tw.F10 GMPEs.*

Description

Variable name of AGA16.tw.F10 GMPEs.

Usage

```
cnames.AGA16.tw.F10
```

Format

Vectors of AGA16.tw.F10 GMPE.

cnames.AM09 *Variable name of AM09 GMPEs.*

Description

Variable name of AM09 GMPEs.

Usage

```
cnames.AM09
```

Format

Vectors of AM09 GMPE.

cnames.Arroyo2010 *Variable name of Arroyo2010 GMPEs.*

Description

Variable name of Arroyo2010 GMPEs.

Usage

```
cnames.Arroyo2010
```

Format

Vectors of Arroyo2010 GMPE.

cnames.ASB14 *Variable name of ASB14 GMPEs.*

Description

Variable name of ASB14 GMPEs.

Usage

```
cnames.ASB14
```

Format

Vectors of ASB14 GMPE.

cnames.ASB14.tw.B01 *Variable name of ASB14.tw.B01 GMPEs.*

Description

Variable name of ASB14.tw.B01 GMPEs.

Usage

```
cnames.ASB14.tw.B01
```

Format

Vectors of ASB14.tw.B01 GMPE.

cnames.ASB14.tw.C01 *Variable name of ASB14.tw.C01 GMPEs.*

Description

Variable name of ASB14.tw.C01 GMPEs.

Usage

```
cnames.ASB14.tw.C01
```

Format

Vectors of ASB14.tw.C01 GMPE.

cnames.ASB14.tw.E02 *Variable name of ASB14.tw.E02 GMPEs.*

Description

Variable name of ASB14.tw.E02 GMPEs.

Usage

```
cnames.ASB14.tw.E02
```

Format

Vectors of ASB14.tw.E02 GMPE.

cnames.ASK14 *Variable name of ASK14 GMPEs.*

Description

Variable name of ASK14 GMPEs.

Usage

```
cnames.ASK14
```

Format

Vectors of ASK14 GMPE.

cnames.ASK14.tw.B01 *Variable name of ASK14.tw.B01 GMPEs.*

Description

Variable name of ASK14.tw.B01 GMPEs.

Usage

```
cnames.ASK14.tw.B01
```

Format

Vectors of ASK14.tw.B01 GMPE.

cnames.ASK14.tw.C01 *Variable name of ASK14.tw.C01 GMPEs.*

Description

Variable name of ASK14.tw.C01 GMPEs.

Usage

```
cnames.ASK14.tw.C01
```

Format

Vectors of ASK14.tw.C01 GMPE.

cnames.ASK14.tw.E03 *Variable name of ASK14.tw.E03 GMPEs.*

Description

Variable name of ASK14.tw.E03 GMPEs.

Usage

```
cnames.ASK14.tw.E03
```

Format

Vectors of ASK14.tw.E03 GMPE.

cnames.BCHydroSub2018Global *Variable name of BCHydroSub2018Global GMPEs.*

Description

Variable name of BCHydroSub2018Global GMPEs.

Usage

```
cnames.BCHydroSub2018Global
```

Format

Vectors of BCHydroSub2018Global GMPE.

cnames.BCHydroSub2018SouthAM *Variable name of BCHydroSub2018SouthAM GMPEs.*

Description

Variable name of BCHydroSub2018SouthAM GMPEs.

Usage

```
cnames.BCHydroSub2018SouthAM
```

Format

Vectors of BCHydroSub2018SouthAM GMPE.

cnames.BCHydroSub2018TW *Variable name of BCHydroSub2018TW GMPEs.*

Description

Variable name of BCHydroSub2018TW GMPEs.

Usage

```
cnames.BCHydroSub2018TW
```

Format

Vectors of BCHydroSub2018TW GMPE.

cnames.BCHydroSubV3 *Variable name of BCHydroSubV3 GMPEs.*

Description

Variable name of BCHydroSubV3 GMPEs.

Usage

```
cnames.BCHydroSubV3
```

Format

Vectors of BCHydroSubV3 GMPE.

cnames.BI14 *Variable name of BI14 GMPEs.*

Description

Variable name of BI14 GMPEs.

Usage

```
cnames.BI14
```

Format

Vectors of BI14 GMPE.

cnames.BI14.tw.B01 *Variable name of BI14.tw.B01 GMPEs.*

Description

Variable name of BI14.tw.B01 GMPEs.

Usage

```
cnames.BI14.tw.B01
```

Format

Vectors of BI14.tw.B01 GMPE.

cnames.BI14.tw.C01 *Variable name of BI14.tw.C01 GMPEs.*

Description

Variable name of BI14.tw.C01 GMPEs.

Usage

```
cnames.BI14.tw.C01
```

Format

Vectors of BI14.tw.C01 GMPE.

cnames.BI14.tw.E02 *Variable name of BI14.tw.E02 GMPEs.*

Description

Variable name of BI14.tw.E02 GMPEs.

Usage

```
cnames.BI14.tw.E02
```

Format

Vectors of BI14.tw.E02 GMPE.

cnames.BSSA14 *Variable name of BSSA14 GMPEs.*

Description

Variable name of BSSA14 GMPEs.

Usage

```
cnames.BSSA14
```

Format

Vectors of BSSA14 GMPE.

cnames.BSSA14.tw.B01 *Variable name of BSSA14.tw.B01 GMPEs.*

Description

Variable name of BSSA14.tw.B01 GMPEs.

Usage

```
cnames.BSSA14.tw.B01
```

Format

Vectors of BSSA14.tw.B01 GMPE.

cnames.BSSA14.tw.C01 *Variable name of BSSA14.tw.C01 GMPEs.*

Description

Variable name of BSSA14.tw.C01 GMPEs.

Usage

```
cnames.BSSA14.tw.C01
```

Format

Vectors of BSSA14.tw.C01 GMPE.

cnames.BSSA14.tw.E02 *Variable name of BSSA14.tw.E02 GMPEs.*

Description

Variable name of BSSA14.tw.E02 GMPEs.

Usage

```
cnames.BSSA14.tw.E02
```

Format

Vectors of BSSA14.tw.E02 GMPE.

cnames.Camp03H *Variable name of Camp03H GMPEs.*

Description

Variable name of Camp03H GMPEs.

Usage

```
cnames.Camp03H
```

Format

Vectors of Camp03H GMPE.

cnames.CB14 *Variable name of CB14 GMPEs.*

Description

Variable name of CB14 GMPEs.

Usage

```
cnames.CB14
```

Format

Vectors of CB14 GMPE.

cnames.CB14.tw.C01 *Variable name of CB14.tw.C01 GMPEs.*

Description

Variable name of CB14.tw.C01 GMPEs.

Usage

```
cnames.CB14.tw.C01
```

Format

Vectors of CB14.tw.C01 GMPE.

cnames.CB14.tw.E05 *Variable name of CB14.tw.E05 GMPEs.*

Description

Variable name of CB14.tw.E05 GMPEs.

Usage

```
cnames.CB14.tw.E05
```

Format

Vectors of CB14.tw.E05 GMPE.

cnames.Chao2017 *Variable name of Chao2017 GMPEs.*

Description

Variable name of Chao2017 GMPEs.

Usage

```
cnames.Chao2017
```

Format

Vectors of Chao2017 GMPE.

cnames.Chao2018 *Variable name of Chao2018 GMPEs.*

Description

Variable name of Chao2018 GMPEs.

Usage

```
cnames.Chao2018
```

Format

Vectors of Chao2018 GMPE.

cnames.Cru.Com.001 *Variable name of Cru.Com.001 GMPEs.*

Description

Variable name of Cru.Com.001 GMPEs.

Usage

```
cnames.Cru.Com.001
```

Format

Vectors of Cru.Com.001 GMPE.

cnames.Cru.Com.002 *Variable name of Cru.Com.002 GMPEs.*

Description

Variable name of Cru.Com.002 GMPEs.

Usage

```
cnames.Cru.Com.002
```

Format

Vectors of Cru.Com.002 GMPE.

cnames.Cru.Com.003 *Variable name of Cru.Com.003 GMPEs.*

Description

Variable name of Cru.Com.003 GMPEs.

Usage

```
cnames.Cru.Com.003
```

Format

Vectors of Cru.Com.003 GMPE.

cnames.Cru.Com.004 *Variable name of Cru.Com.004 GMPEs.*

Description

Variable name of Cru.Com.004 GMPEs.

Usage

```
cnames.Cru.Com.004
```

Format

Vectors of Cru.Com.004 GMPE.

cnames.Cru.Com.005 *Variable name of Cru.Com.005 GMPEs.*

Description

Variable name of Cru.Com.005 GMPEs.

Usage

```
cnames.Cru.Com.005
```

Format

Vectors of Cru.Com.005 GMPE.

cnames.Cru.Com.006 *Variable name of Cru.Com.006 GMPEs.*

Description

Variable name of Cru.Com.006 GMPEs.

Usage

```
cnames.Cru.Com.006
```

Format

Vectors of Cru.Com.006 GMPE.

cnames.Cru.Com.007 *Variable name of Cru.Com.007 GMPEs.*

Description

Variable name of Cru.Com.007 GMPEs.

Usage

```
cnames.Cru.Com.007
```

Format

Vectors of Cru.Com.007 GMPE.

cnames.Cru.Com.008 *Variable name of Cru.Com.008 GMPEs.*

Description

Variable name of Cru.Com.008 GMPEs.

Usage

```
cnames.Cru.Com.008
```

Format

Vectors of Cru.Com.008 GMPE.

cnames.Cru.Com.009 *Variable name of Cru.Com.009 GMPEs.*

Description

Variable name of Cru.Com.009 GMPEs.

Usage

```
cnames.Cru.Com.009
```

Format

Vectors of Cru.Com.009 GMPE.

cnames.Cru.Com.010 *Variable name of Cru.Com.010 GMPEs.*

Description

Variable name of Cru.Com.010 GMPEs.

Usage

```
cnames.Cru.Com.010
```

Format

Vectors of Cru.Com.010 GMPE.

cnames.Cru.Com.011 *Variable name of Cru.Com.011 GMPEs.*

Description

Variable name of Cru.Com.011 GMPEs.

Usage

```
cnames.Cru.Com.011
```

Format

Vectors of Cru.Com.011 GMPE.

cnames.Cru.Com.012 *Variable name of Cru.Com.012 GMPEs.*

Description

Variable name of Cru.Com.012 GMPEs.

Usage

```
cnames.Cru.Com.012
```

Format

Vectors of Cru.Com.012 GMPE.

cnames.Cru.Com.013 *Variable name of Cru.Com.013 GMPEs.*

Description

Variable name of Cru.Com.013 GMPEs.

Usage

```
cnames.Cru.Com.013
```

Format

Vectors of Cru.Com.013 GMPE.

cnames.Cru.Com.014 *Variable name of Cru.Com.014 GMPEs.*

Description

Variable name of Cru.Com.014 GMPEs.

Usage

```
cnames.Cru.Com.014
```

Format

Vectors of Cru.Com.014 GMPE.

cnames.Cru.Com.015 *Variable name of Cru.Com.015 GMPEs.*

Description

Variable name of Cru.Com.015 GMPEs.

Usage

```
cnames.Cru.Com.015
```

Format

Vectors of Cru.Com.015 GMPE.

cnames.Cru.Com.016 *Variable name of Cru.Com.016 GMPEs.*

Description

Variable name of Cru.Com.016 GMPEs.

Usage

```
cnames.Cru.Com.016
```

Format

Vectors of Cru.Com.016 GMPE.

cnames.Cru.Com.017 *Variable name of Cru.Com.017 GMPEs.*

Description

Variable name of Cru.Com.017 GMPEs.

Usage

```
cnames.Cru.Com.017
```

Format

Vectors of Cru.Com.017 GMPE.

cnames.CY08 *Variable name of CY08 GMPEs.*

Description

Variable name of CY08 GMPEs.

Usage

```
cnames.CY08
```

Format

Vectors of CY08 GMPE.

cnames.CY14 *Variable name of CY14 GMPEs.*

Description

Variable name of CY14 GMPEs.

Usage

```
cnames.CY14
```

Format

Vectors of CY14 GMPE.

cnames.CY14.tw.B01 *Variable name of CY14.tw.B0 GMPEs.*

Description

Variable name of CY14.tw.B0 GMPEs.

Usage

```
cnames.CY14.tw.B01
```

Format

Vectors of CY14.tw.B0 GMPE.

cnames.CY14.tw.C01 *Variable name of CY14.tw.C01 GMPEs.*

Description

Variable name of CY14.tw.C01 GMPEs.

Usage

```
cnames.CY14.tw.C01
```

Format

Vectors of CY14.tw.C01 GMPE.

cnames.CY14.tw.E04 *Variable name of CY14.tw.E04 GMPEs.*

Description

Variable name of CY14.tw.E04 GMPEs.

Usage

```
cnames.CY14.tw.E04
```

Format

Vectors of CY14.tw.E04 GMPE.

cnames.Garcia05 *Variable name of Garcia05 GMPEs.*

Description

Variable name of Garcia05 GMPEs.

Usage

```
cnames.Garcia05
```

Format

Vectors of Garcia05 GMPE.

cnames.GK16 *Variable name of GK16 GMPEs.*

Description

Variable name of GK16 GMPEs.

Usage

```
cnames.GK16
```

Format

Vectors of GK16 GMPE.

cnames.Gregor06 *Variable name of Gregor06 GMPEs.*

Description

Variable name of Gregor06 GMPEs.

Usage

```
cnames.Gregor06
```

Format

Vectors of Gregor06 GMPE.

cnames.I14 *Variable name of I14 GMPEs.*

Description

Variable name of I14 GMPEs.

Usage

```
cnames.I14
```

Format

Vectors of I14 GMPE.

cnames.I14.tw.B01 *Variable name of I14.tw.B01 GMPEs.*

Description

Variable name of I14.tw.B01 GMPEs.

Usage

```
cnames.I14.tw.B01
```

Format

Vectors of I14.tw.B01 GMPE.

cnames.I14.tw.C01 *Variable name of I14.tw.C01 GMPEs.*

Description

Variable name of I14.tw.C01 GMPEs.

Usage

```
cnames.I14.tw.C01
```

Format

Vectors of I14.tw.C01 GMPE.

cnames.I14.tw.E04 *Variable name of I14.tw.E04 GMPEs.*

Description

Variable name of I14.tw.E04 GMPEs.

Usage

```
cnames.I14.tw.E04
```

Format

Vectors of I14.tw.E04 GMPE.

cnames.KAAH15 *Variable name of KAAH15 GMPEs.*

Description

Variable name of KAAH15 GMPEs.

Usage

```
cnames.KAAH15
```

Format

Vectors of KAAH15 GMPE.

cnames.Kanno06 *Variable name of Kanno06 GMPEs.*

Description

Variable name of Kanno06 GMPEs.

Usage

```
cnames.Kanno06
```

Format

Vectors of Kanno06 GMPE.

cnames.Lin09 *Variable name of Lin09 GMPEs.*

Description

Variable name of Lin09 GMPEs.

Usage

```
cnames.Lin09
```

Format

Vectors of Lin09 GMPE.

cnames.Lin11 *Variable name of Lin11 GMPEs.*

Description

Variable name of Lin11 GMPEs.

Usage

```
cnames.Lin11
```

Format

Vectors of Lin11 GMPE.

cnames.Lin11Rock *Variable name of Lin11Rock GMPEs.*

Description

Variable name of Lin11Rock GMPEs.

Usage

```
cnames.Lin11Rock
```

Format

Vectors of Lin11Rock GMPE.

cnames.Lin11Soil *Variable name of Lin11Soil GMPEs.*

Description

Variable name of Lin11Soil GMPEs.

Usage

```
cnames.Lin11Soil
```

Format

Vectors of Lin11Soil GMPE.

cnames.LL08 *Variable name of LL08 GMPEs.*

Description

Variable name of LL08 GMPEs.

Usage

```
cnames.LL08
```

Format

Vectors of LL08 GMPE.

cnames.LL08.C02 *Variable name of LL08.C02 GMPEs.*

Description

Variable name of LL08.C02 GMPEs.

Usage

```
cnames.LL08.C02
```

Format

Vectors of LL08.C02 GMPE.

cnames.LL08.E04 *Variable name of LL08.E04 GMPEs.*

Description

Variable name of LL08.E04 GMPEs.

Usage

```
cnames.LL08.E04
```

Format

Vectors of LL08.E04 GMPE.

cnames.LL08.F04 *Variable name of LL08.F04 GMPEs.*

Description

Variable name of LL08.F04 GMPEs.

Usage

```
cnames.LL08.F04
```

Format

Vectors of LL08.F04 GMPE.

cnames.LL08Rock *Variable name of LL08Rock GMPEs.*

Description

Variable name of LL08Rock GMPEs.

Usage

```
cnames.LL08Rock
```

Format

Vectors of LL08Rock GMPE.

cnames.LL08Soil *Variable name of LL08Soil GMPEs.*

Description

Variable name of LL08Soil GMPEs.

Usage

```
cnames.LL08Soil
```

Format

Vectors of LL08Soil GMPE.

cnames.Loh96 *Variable name of Loh96 GMPEs.*

Description

Variable name of Loh96 GMPEs.

Usage

```
cnames.Loh96
```

Format

Vectors of Loh96 GMPE.

cnames.MBR17 *Variable name of MBR17 GMPEs.*

Description

Variable name of MBR17 GMPEs.

Usage

```
cnames.MBR17
```

Format

Vectors of MBR17 GMPE.

cnames.Mcverry2006Sub *Variable name of Mcverry2006Sub GMPEs.*

Description

Variable name of Mcverry2006Sub GMPEs.

Usage

```
cnames.Mcverry2006Sub
```

Format

Vectors of Mcverry2006Sub GMPE.

cnames.NCREE2011 *Variable name of NCREE2011 GMPEs.*

Description

Variable name of NCREE2011 GMPEs.

Usage

```
cnames.NCREE2011
```

Format

Vectors of NCREE2011 GMPE.

cnames.PhungCru17 *Variable name of PhungCru17 GMPEs.*

Description

Variable name of PhungCru17 GMPEs.

Usage

```
cnames.PhungCru17
```

Format

Vectors of PhungCru17 GMPE.

cnames.PhungCru18 *Variable name of PhungCru18 GMPEs.*

Description

Variable name of PhungCru18 GMPEs.

Usage

```
cnames.PhungCru18
```

Format

Vectors of PhungCru18 GMPE.

cnames.PhungSub17 *Variable name of PhungSub17 GMPEs.*

Description

Variable name of PhungSub17 GMPEs.

Usage

```
cnames.PhungSub17
```

Format

Vectors of PhungSub17 GMPE.

cnames.PhungSub18 *Variable name of PhungSub18 GMPEs.*

Description

Variable name of PhungSub18 GMPEs.

Usage

```
cnames.PhungSub18
```

Format

Vectors of PhungSub18 GMPE.

cnames.Sinotech12 *Variable name of Sinotech12 GMPEs.*

Description

Variable name of Sinotech12 GMPEs.

Usage

```
cnames.Sinotech12
```

Format

Vectors of Sinotech12 GMPE.

cnames.Sub.Inter.Com.001 *Variable name of Sub.Inter.Com.001 GMPEs.*

Description

Variable name of Sub.Inter.Com.001 GMPEs.

Usage

```
cnames.Sub.Inter.Com.001
```

Format

Vectors of Sub.Inter.Com.001 GMPE.

cnames.Sub.Inter.Com.002 *Variable name of Sub.Inter.Com.002 GMPEs.*

Description

Variable name of Sub.Inter.Com.002 GMPEs.

Usage

```
cnames.Sub.Inter.Com.002
```

Format

Vectors of Sub.Inter.Com.002 GMPE.

`cnames.Sub.Inter.Com.003`

Variable name of Sub.Inter.Com.003 GMPEs.

Description

Variable name of Sub.Inter.Com.003 GMPEs.

Usage

```
cnames.Sub.Inter.Com.003
```

Format

Vectors of Sub.Inter.Com.003 GMPE.

`cnames.Sub.Inter.Com.004`

Variable name of Sub.Inter.Com.004 GMPEs.

Description

Variable name of Sub.Inter.Com.004 GMPEs.

Usage

```
cnames.Sub.Inter.Com.004
```

Format

Vectors of Sub.Inter.Com.004 GMPE.

`cnames.Sub.Inter.Com.005`

Variable name of Sub.Inter.Com.005 GMPEs.

Description

Variable name of Sub.Inter.Com.005 GMPEs.

Usage

```
cnames.Sub.Inter.Com.005
```

Format

Vectors of Sub.Inter.Com.005 GMPE.

`cnames.Sub.Inter.Com.006`
Variable name of Sub.Inter.Com.006 GMPEs.

Description

Variable name of Sub.Inter.Com.006 GMPEs.

Usage

```
cnames.Sub.Inter.Com.006
```

Format

Vectors of Sub.Inter.Com.006 GMPE.

`cnames.Sub.Inter.Com.007`
Variable name of Sub.Inter.Com.007 GMPEs.

Description

Variable name of Sub.Inter.Com.007 GMPEs.

Usage

```
cnames.Sub.Inter.Com.007
```

Format

Vectors of Sub.Inter.Com.007 GMPE.

`cnames.Sub.Inter.Com.008`
Variable name of Sub.Inter.Com.008 GMPEs.

Description

Variable name of Sub.Inter.Com.008 GMPEs.

Usage

```
cnames.Sub.Inter.Com.008
```

Format

Vectors of Sub.Inter.Com.008 GMPE.

`cnames.Sub.Inter.Com.009`
Variable name of Sub.Inter.Com.009 GMPEs.

Description

Variable name of Sub.Inter.Com.009 GMPEs.

Usage

```
cnames.Sub.Inter.Com.009
```

Format

Vectors of Sub.Inter.Com.009 GMPE.

`cnames.Sub.Inter.Com.010`
Variable name of Sub.Inter.Com.010 GMPEs.

Description

Variable name of Sub.Inter.Com.010 GMPEs.

Usage

```
cnames.Sub.Inter.Com.010
```

Format

Vectors of Sub.Inter.Com.010 GMPE.

`cnames.Sub.Inter.Com.011`
Variable name of Sub.Inter.Com.011 GMPEs.

Description

Variable name of Sub.Inter.Com.011 GMPEs.

Usage

```
cnames.Sub.Inter.Com.011
```

Format

Vectors of Sub.Inter.Com.011 GMPE.

`cnames.Sub.Inter.Com.012` *Variable name of Sub.Inter.Com.012 GMPEs.*

Description

Variable name of Sub.Inter.Com.012 GMPEs.

Usage

```
cnames.Sub.Inter.Com.012
```

Format

Vectors of Sub.Inter.Com.012 GMPE.

`cnames.Sub.Inter.Com.013` *Variable name of Sub.Inter.Com.013 GMPEs.*

Description

Variable name of Sub.Inter.Com.013 GMPEs.

Usage

```
cnames.Sub.Inter.Com.013
```

Format

Vectors of Sub.Inter.Com.013 GMPE.

`cnames.Sub.Inter.Com.014` *Variable name of Sub.Inter.Com.014 GMPEs.*

Description

Variable name of Sub.Inter.Com.014 GMPEs.

Usage

```
cnames.Sub.Inter.Com.014
```

Format

Vectors of Sub.Inter.Com.014 GMPE.

`cnames.Sub.Inter.Com.015`

Variable name of Sub.Inter.Com.015 GMPEs.

Description

Variable name of Sub.Inter.Com.015 GMPEs.

Usage

```
cnames.Sub.Inter.Com.015
```

Format

Vectors of Sub.Inter.Com.015 GMPE.

`cnames.Sub.Inter.Com.016`

Variable name of Sub.Inter.Com.016 GMPEs.

Description

Variable name of Sub.Inter.Com.016 GMPEs.

Usage

```
cnames.Sub.Inter.Com.016
```

Format

Vectors of Sub.Inter.Com.016 GMPE.

`cnames.Sub.Inter.Com.017`

Variable name of Sub.Inter.Com.017 GMPEs.

Description

Variable name of Sub.Inter.Com.017 GMPEs.

Usage

```
cnames.Sub.Inter.Com.017
```

Format

Vectors of Sub.Inter.Com.017 GMPE.

`cnames.Sub.Intra.Com.001`
Variable name of Sub.Intra.Com.001 GMPEs.

Description

Variable name of Sub.Intra.Com.001 GMPEs.

Usage

```
cnames.Sub.Intra.Com.001
```

Format

Vectors of Sub.Intra.Com.001 GMPE.

`cnames.Sub.Intra.Com.002`
Variable name of Sub.Intra.Com.002 GMPEs.

Description

Variable name of Sub.Intra.Com.002 GMPEs.

Usage

```
cnames.Sub.Intra.Com.002
```

Format

Vectors of Sub.Intra.Com.002 GMPE.

`cnames.Sub.Intra.Com.003`
Variable name of Sub.Intra.Com.003 GMPEs.

Description

Variable name of Sub.Intra.Com.003 GMPEs.

Usage

```
cnames.Sub.Intra.Com.003
```

Format

Vectors of Sub.Intra.Com.003 GMPE.

`cnames.Sub.Intra.Com.004`

Variable name of Sub.Intra.Com.004 GMPEs.

Description

Variable name of Sub.Intra.Com.004 GMPEs.

Usage

```
cnames.Sub.Intra.Com.004
```

Format

Vectors of Sub.Intra.Com.004 GMPE.

`cnames.Sub.Intra.Com.005`

Variable name of Sub.Intra.Com.005 GMPEs.

Description

Variable name of Sub.Intra.Com.005 GMPEs.

Usage

```
cnames.Sub.Intra.Com.005
```

Format

Vectors of Sub.Intra.Com.005 GMPE.

`cnames.Sub.Intra.Com.006`

Variable name of Sub.Intra.Com.006 GMPEs.

Description

Variable name of Sub.Intra.Com.006 GMPEs.

Usage

```
cnames.Sub.Intra.Com.006
```

Format

Vectors of Sub.Intra.Com.006 GMPE.

`cnames.Sub.Intra.Com.007`
Variable name of Sub.Intra.Com.007 GMPEs.

Description

Variable name of Sub.Intra.Com.007 GMPEs.

Usage

```
cnames.Sub.Intra.Com.007
```

Format

Vectors of Sub.Intra.Com.007 GMPE.

`cnames.Sub.Intra.Com.008`
Variable name of Sub.Intra.Com.008 GMPEs.

Description

Variable name of Sub.Intra.Com.008 GMPEs.

Usage

```
cnames.Sub.Intra.Com.008
```

Format

Vectors of Sub.Intra.Com.008 GMPE.

`cnames.Sub.Intra.Com.009`
Variable name of Sub.Intra.Com.009 GMPEs.

Description

Variable name of Sub.Intra.Com.009 GMPEs.

Usage

```
cnames.Sub.Intra.Com.009
```

Format

Vectors of Sub.Intra.Com.009 GMPE.

`cnames.Sub.Intra.Com.010`

Variable name of Sub.Intra.Com.010 GMPEs.

Description

Variable name of Sub.Intra.Com.010 GMPEs.

Usage

```
cnames.Sub.Intra.Com.010
```

Format

Vectors of Sub.Intra.Com.010 GMPE.

`cnames.Sub.Intra.Com.011`

Variable name of Sub.Intra.Com.011 GMPEs.

Description

Variable name of Sub.Intra.Com.011 GMPEs.

Usage

```
cnames.Sub.Intra.Com.011
```

Format

Vectors of Sub.Intra.Com.011 GMPE.

`cnames.Sub.Intra.Com.012`

Variable name of Sub.Intra.Com.012 GMPEs.

Description

Variable name of Sub.Intra.Com.012 GMPEs.

Usage

```
cnames.Sub.Intra.Com.012
```

Format

Vectors of Sub.Intra.Com.012 GMPE.

`cnames.Sub.Intra.Com.013`

Variable name of Sub.Intra.Com.013 GMPEs.

Description

Variable name of Sub.Intra.Com.013 GMPEs.

Usage

```
cnames.Sub.Intra.Com.013
```

Format

Vectors of Sub.Intra.Com.013 GMPE.

`cnames.Sub.Intra.Com.014`

Variable name of Sub.Intra.Com.014 GMPEs.

Description

Variable name of Sub.Intra.Com.014 GMPEs.

Usage

```
cnames.Sub.Intra.Com.014
```

Format

Vectors of Sub.Intra.Com.014 GMPE.

`cnames.Sub.Intra.Com.015`

Variable name of Sub.Intra.Com.015 GMPEs.

Description

Variable name of Sub.Intra.Com.015 GMPEs.

Usage

```
cnames.Sub.Intra.Com.015
```

Format

Vectors of Sub.Intra.Com.015 GMPE.

cnames.Sub.Intra.Com.016

Variable name of Sub.Intra.Com.016 GMPEs.

Description

Variable name of Sub.Intra.Com.016 GMPEs.

Usage

```
cnames.Sub.Intra.Com.016
```

Format

Vectors of Sub.Intra.Com.016 GMPE.

cnames.Sub.Intra.Com.017

Variable name of Sub.Intra.Com.017 GMPEs.

Description

Variable name of Sub.Intra.Com.017 GMPEs.

Usage

```
cnames.Sub.Intra.Com.017
```

Format

Vectors of Sub.Intra.Com.017 GMPE.

cnames.Youngs97Rock

Variable name of Youngs97Rock GMPEs.

Description

Variable name of Youngs97Rock GMPEs.

Usage

```
cnames.Youngs97Rock
```

Format

Vectors of Youngs97Rock GMPE.

cnames.Zh06 *Variable name of Zh06 GMPEs.*

Description

Variable name of Zh06 GMPEs.

Usage

```
cnames.Zh06
```

Format

Vectors of Zh06 GMPE.

cnames.Zh16Cru *Variable name of Zh16Cru GMPEs.*

Description

Variable name of Zh16Cru GMPEs.

Usage

```
cnames.Zh16Cru
```

Format

Vectors of Zh16Cru GMPE.

cnames.Zh16Inter *Variable name of Zh16Inter GMPEs.*

Description

Variable name of Zh16Inter GMPEs.

Usage

```
cnames.Zh16Inter
```

Format

Vectors of Zh16Inter GMPE.

cnames.Zh16Slab *Variable name of Zh16Slab GMPEs.*

Description

Variable name of Zh16Slab GMPEs.

Usage

```
cnames.Zh16Slab
```

Format

Vectors of Zh16Slab GMPE.

cnames.Zh16Sub *Variable name of Zh16Sub GMPEs.*

Description

Variable name of Zh16Sub GMPEs.

Usage

```
cnames.Zh16Sub
```

Format

Vectors of Zh16Sub GMPE.

Cru.Com.001 *GMPE function for Crustal Common form 001 (2017)*

Description

`Cru.Com.001` returns the ground-motion prediction with it sigma of Crustal Common form 001 GMPE.

Usage

```
Cru.Com.001(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 001

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.001(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.001(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.002 *GMPE function for Crustal Common form 002 (2017)*

Description

Cru.Com.002 returns the ground-motion prediction with it sigma of Crustal Common form 002 GMPE.

Usage

```
Cru.Com.002(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 002

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.002(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.002(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

`Cru.Com.003` *GMPE function for Crustal Common form 003 (2017)*

Description

`Cru.Com.003` returns the ground-motion prediction with it sigma of Crustal Common form 003 GMPE.

Usage

```
Cru.Com.003(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`ftype`	style of faulting.
`Dip`	Dip angle of the fault plane.
`rupwidth`	Down-dip rupture width (km).
`Rx`	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
`hwflag`	hanging-wall flag, 1 for hanging-wall.
`Prd`	Period of spectral acceleration.

Details

Crustal Common form 003

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.003(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.003(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.004 *GMPE function for Crustal Common form 004 (2017)*

Description

Cru.Com.004 returns the ground-motion prediction with it sigma of Crustal Common form 004 GMPE.

Usage

```
Cru.Com.004(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 004

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.004(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.004(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

`Cru.Com.005` *GMPE function for Crustal Common form 005 (2017)*

Description

`Cru.Com.005` returns the ground-motion prediction with it sigma of Crustal Common form 005 GMPE.

Usage

```
Cru.Com.005(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`ftype`	style of faulting.
`Dip`	Dip angle of the fault plane.
`rupwidth`	Down-dip rupture width (km).
`Rx`	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
`hwflag`	hanging-wall flag, 1 for hanging-wall.
`Prd`	Period of spectral acceleration.

Details

Crustal Common form 005

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.005(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.005(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.006 *GMPE function for Crustal Common form 006 (2017)*

Description

Cru.Com.006 returns the ground-motion prediction with it sigma of Crustal Common form 006 GMPE.

Usage

```
Cru.Com.006(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 006

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.006(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.006(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

`Cru.Com.007` *GMPE function for Crustal Common form 007 (2017)*

Description

`Cru.Com.007` returns the ground-motion prediction with it sigma of Crustal Common form 007 GMPE.

Usage

```
Cru.Com.007(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`ftype`	style of faulting.
`Dip`	Dip angle of the fault plane.
`rupwidth`	Down-dip rupture width (km).
`Rx`	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
`hwflag`	hanging-wall flag, 1 for hanging-wall.
`Prd`	Period of spectral acceleration.

Details

Crustal Common form 007

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.007(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.007(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.008 *GMPE function for Crustal Common form 008 (2017)*

Description

Cru.Com.008 returns the ground-motion prediction with it sigma of Crustal Common form 008 GMPE.

Usage

```
Cru.Com.008(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 008

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.008(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.008(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.009 *GMPE function for Crustal Common form 009 (2017)*

Description

Cru.Com.009 returns the ground-motion prediction with it sigma of Crustal Common form 009 GMPE.

Usage

```
Cru.Com.009(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 009

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.009(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.009(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.010 *GMPE function for Crustal Common form 010 (2017)*

Description

Cru.Com.010 returns the ground-motion prediction with it sigma of Crustal Common form 010 GMPE.

Usage

```
Cru.Com.010(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 010

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.010(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.010(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.011 *GMPE function for Crustal Common form 011 (2017)*

Description

Cru.Com.011 returns the ground-motion prediction with it sigma of Crustal Common form 011 GMPE.

Usage

```
Cru.Com.011(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 011

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.011(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.011(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.012 *GMPE function for Crustal Common form 012 (2017)*

Description

Cru.Com.012 returns the ground-motion prediction with it sigma of Crustal Common form 012 GMPE.

Usage

```
Cru.Com.012(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 012

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.012(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.012(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

`Cru.Com.013` *GMPE function for Crustal Common form 013 (2017)*

Description

`Cru.Com.013` returns the ground-motion prediction with it sigma of Crustal Common form 013 GMPE.

Usage

```
Cru.Com.013(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`ftype`	style of faulting.
`Dip`	Dip angle of the fault plane.
`rupwidth`	Down-dip rupture width (km).
`Rx`	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
`hwflag`	hanging-wall flag, 1 for hanging-wall.
`Prd`	Period of spectral acceleration.

Details

Crustal Common form 013

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.013(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.013(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.014 *GMPE function for Crustal Common form 014 (2017)*

Description

Cru.Com.014 returns the ground-motion prediction with it sigma of Crustal Common form 014 GMPE.

Usage

```
Cru.Com.014(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 014

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.014(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.014(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.015 *GMPE function for Crustal Common form 015 (2017)*

Description

Cru.Com.015 returns the ground-motion prediction with it sigma of Crustal Common form 015 GMPE.

Usage

```
Cru.Com.015(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 015

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.015(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.015(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

Cru.Com.016 *GMPE function for Crustal Common form 016 (2017)*

Description

Cru.Com.016 returns the ground-motion prediction with it sigma of Crustal Common form 016 GMPE.

Usage

```
Cru.Com.016(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Dip	Dip angle of the fault plane.
rupwidth	Down-dip rupture width (km).
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
hwflag	hanging-wall flag, 1 for hanging-wall.
Prd	Period of spectral acceleration.

Details

Crustal Common form 016

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.016(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.016(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

`Cru.Com.017` *GMPE function for Crustal Common form 017 (2017)*

Description

`Cru.Com.017` returns the ground-motion prediction with it sigma of Crustal Common form 017 GMPE.

Usage

```
Cru.Com.017(
  Mag,
  Rrup,
  Rjb,
  Ztor,
  ftype = 0,
  Dip,
  rupwidth,
  Rx,
  hwflag = 0,
  Prd
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`ftype`	style of faulting.
`Dip`	Dip angle of the fault plane.
`rupwidth`	Down-dip rupture width (km).
`Rx`	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
`hwflag`	hanging-wall flag, 1 for hanging-wall.
`Prd`	Period of spectral acceleration.

Details

Crustal Common form 017

Value

A list will be return, including mag, Rrup, Rjb, Ztor, ftype, dip, rupwidth, Rx, hwflag, pecT, lnY, sigma, iflag.

Examples

```
Cru.Com.017(6, 20, 20, 5, 0, 90, 20, 20, 0, 0)
Cru.Com.017(7, 20, 20, 2, 0, 90, 20, 20, 0, 0 )
```

CY08 *GMPE function for Chiou and Youngs(2008) NGA model*

Description

CY08 returns the ground-motion prediction with it sigma of Chiou and Youngs(2008) GMPE.

Usage

```
CY08(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Vs30,
  Dip,
  Ztor,
  ftype = 0,
  Z1.0,
  Vs30_class = 1,
  hwflag = 0,
  Rx,
  regionflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Dip	Dip angle of the fault plane.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Vs30_class	1 measured, 0 estimated.
hwflag	hanging-wall flag, 1 for hanging-wall.
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
regionflag	0 Global, 1 Japan and Italy, 2 Wenchuan (note only for M7.9)

Details

BrianS-J. Chiou and Robert R. Youngs (2008) An NGA Model for the Average Horizontal Component of Peak Ground Motion and Response Spectra. Earthquake Spectra, Vol. 24, No. 1, pp. 173-215. doi:10.1193/1.2894832

Value

A list will be return, including mag, Rrup, Rjb, specT, period, lnY, sigma, iflag, Vs30, dip, Ztor, ftype, Z10, vs3-_class, hwflag, Rx, regionflag, phi, tau.

Examples

```
CY08(6, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
CY08(7, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
```

CY14 *GMPE function for Chiou and Youngs(2014) NGA-West2 model*

Description

CY14 returns the ground-motion prediction with it sigma of Chiou and Youngs(2014) GMPE.

Usage

```
CY14(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Vs30,
  Dip,
  Ztor,
  ftype = 0,
  Z1.0,
  Vs30_class = 1,
  hwflag = 0,
  Rx,
  regionflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Dip	Dip angle of the fault plane.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Vs30_class	1 measured, 0 estimated.
hwflag	hanging-wall flag, 1 for hanging-wall.

Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
regionflag	0 Global, 1 Japan and Italy, 2 Wenchuan (note only for M7.9)

Details

Chiou, B. S.-J., and R. R. Youngs (2014), Update of the chiou and youngs NGA model for the average horizontal component of peak ground motion and response spectra, Earthquake Spectra, 30(3), 1117-1153. doi:10.1193/072813EQS219M

Value

A list will be return, including mag, Rrup, Rjb, specT, period, lnY, sigma, iflag, Vs30, dip, Ztor, ftype, Z10, vs30_class, hwflag, Rx, regionflag, phi, tau.

Examples

```
CY14(6, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
CY14(7, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
```

CY14.tw.B01 *GMPE function for Chiou and Youngs(2014) NGA-West2 model*

Description

CY14.tw.B01 returns the ground-motion prediction with it sigma of Chiou and Youngs(2014) GMPE adjusted to Taiwan.

Usage

```
CY14.tw.B01(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Vs30,
  Dip,
  Ztor,
  ftype = 0,
  Z1.0,
  Vs30_class = 1,
  hwflag = 0,
  Rx,
  regionflag = 0
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Rjb`	Joyner and Boore distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`Vs30`	Vs30(m/s).
`Dip`	Dip angle of the fault plane.
`Ztor`	Depth to the top of the finite rupture model (km).
`ftype`	style of faulting.
`Z1.0`	the depth to the shear-wave velocity horizon of 1.0(km/s).
`Vs30_class`	1 measured, 0 estimated.
`hwflag`	hanging-wall flag, 1 for hanging-wall.
`Rx`	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
`regionflag`	0 Global, 1 Japan and Italy, 2 Wenchuan (note only for M7.9)

Details

Chiou, B. S.-J., and R. R. Youngs (2014), Update of the chiou and youngs NGA model for the average horizontal component of peak ground motion and response spectra, Earthquake Spectra, 30(3), 1117-1153. doi:10.1193/072813EQS219M

Value

A list will be return, including mag, Rrup, Rjb, specT, period, lnY, sigma, iflag, Vs30, dip, Ztor, ftype, Z10, vs3-_class, hwflag, Rx, regionflag, phi, tau.

Examples

```
CY14.tw.B01(6, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
CY14.tw.B01(7, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
```

`CY14.tw.C01` *GMPE function for Chiou and Youngs(2014) NGA-West2 model*

Description

`CY14.tw.C01` returns the ground-motion prediction with it sigma of Chiou and Youngs(2014) GMPE adjusted to Taiwan.

Usage

```
CY14.tw.C01(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Vs30,
  Dip,
  Ztor,
  ftype = 0,
  Z1.0,
  Vs30_class = 1,
  hwflag = 0,
  Rx,
  regionflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Dip	Dip angle of the fault plane.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Vs30_class	1 measured, 0 estimated.
hwflag	hanging-wall flag, 1 for hanging-wall.
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
regionflag	0 Global, 1 Japan and Italy, 2 Wenchuan (note only for M7.9)

Details

Chiou, B. S.-J., and R. R. Youngs (2014), Update of the chiou and youngs NGA model for the average horizontal component of peak ground motion and response spectra, Earthquake Spectra, 30(3), 1117-1153. doi:10.1193/072813EQS219M

Value

A list will be return, including mag, Rrup, Rjb, specT, period, lnY, sigma, iflag, Vs30, dip, Ztor, ftype, Z10, vs3-_class, hwflag, Rx, regionflag, phi, tau.

Examples

```
CY14.tw.C01(6, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
CY14.tw.C01(7, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
```

CY14.tw.E04 *GMPE function for Chiou and Youngs(2014) NGA-West2 model*

Description

CY14.tw.E04 returns the ground-motion prediction with it sigma of Chiou and Youngs(2014) GMPE adjusted to Taiwan.

Usage

```
CY14.tw.E04(
  Mag,
  Rrup,
  Rjb,
  Prd,
  Vs30,
  Dip,
  Ztor,
  ftype = 0,
  Z1.0,
  Vs30_class = 1,
  hwflag = 0,
  Rx,
  regionflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Dip	Dip angle of the fault plane.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Vs30_class	1 measured, 0 estimated.
hwflag	hanging-wall flag, 1 for hanging-wall.
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
regionflag	0 Global, 1 Japan and Italy, 2 Wenchuan (note only for M7.9)

Details

Chiou, B. S.-J., and R. R. Youngs (2014), Update of the chiou and youngs NGA model for the average horizontal component of peak ground motion and response spectra, Earthquake Spectra, 30(3), 1117-1153. doi:10.1193/072813EQS219M

Value

A list will be return, including mag, Rrup, Rjb, specT, period, lnY, sigma, iflag, Vs30, dip, Ztor, ftype, Z10, vs3-_class, hwflag, Rx, regionflag, phi, tau.

Examples

```
CY14.tw.E04(6, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
CY14.tw.E04(7, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 0, 20, 0)
```

Garcia05 *GMPE function for Garcia et al.(2005)*

Description

`Garcia05` returns the ground-motion prediction with it sigma of Garcia et al.(2005) GMPE.

Usage

```
Garcia05(Mag, Rrup, Prd, depth)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`depth`	hypocentral depth(km).

Details

Garcia, D., Singh, S. K., Herraiz, M., Ordaz, M., &Pacheco, J. F. (2005). Inslab Earthquakes of Central Mexico: Peak Ground-Motion Parameters and Response Spectra. Bulletin of the Seismological Society of America, 95(6), 2272-2282.

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag, depth.

Examples

```
Garcia05(6, 20, 0, 35)
Garcia05(7, 10, 0, 60)
```

get.GMPE.prd *To obtain usable period range of GMPE*

Description

get.GMPE.prd returns the the minimum and maximum period of any GMPE.

Usage

```
get.GMPE.prd(GMPE)
```

Arguments

GMPE	Name of GMPE.

Value

A data.frame of period.min and period.max will be return.

Examples

```
get.GMPE.prd("ASK14")
get.GMPE.prd("CY14")
```

GK16 *GMPE function for Graizer and Kalkan(2016)*

Description

GK16 returns the ground-motion prediction with it sigma of Graizer and Kalkan(2016) GMPE.

Usage

```
GK16(Mag, Rrup, ftype = 0, Prd, Vs30 = 760, q0 = 150, Z1.5)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
ftype	style of faulting.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
q0	Quality factor (Q0).
Z1.5	Depth to the 1.5km/s shear-wave velocity isosurfac.

Details

Graizer and E. Kalkan.(2016) Summary of the GK15 ground-motion prediction equation for horizontal PGA and 5 Bulletin of the Seismological Society of America, 106(2), 687-707. doi:10.1785/0120150194

Value

A list will be return, including mag, Rrup, specT, ftype, period, lnY, sigma, iflag, Vs30, q0, Z15.

Examples

```
GK16(6, 20, 0, 0, 760, 150, 1)
GK16(7, 10, 0, 0, 760, 150, 1)
```

GMPEhaz *GMPEhaz: A package for wrapping GMPE functions from Norman Abrahamson's fortran HAZ code.*

Description

The GMPEhaz package provides several GMPE functions: AB03, AC10, ASB13, ASK14, BCHydroSubV3, BI14, BSSA14, CB14, CY14, Gregor06, Kanno06, Lin11, LL08, Youngs97, Zh06.

Usage

```
.onUnload(libpath)
```

Arguments

libpath	library path

GMPEhaz.model *GMPE model list.*

Description

To obtain useable period range for each GMPE model.

Usage

```
GMPEhaz.model
```

Format

dataframe of GMPE model list.

Gregor06 *GMPE function for Gregor et al.(2006)*

Description

Gregor06 returns the ground-motion prediction with it sigma of Gregor et al.(2006) GMPE.

Usage

```
Gregor06(Mag, Rrup, Prd, Vs30)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).

Details

Gregor, N., W. Silva, I. Wong, and R. Youngs (2006), Updated response spectral attenuation relationship for cascadia subduction zone megathrust earthquake, Seismological Research Letters, 77(2), 325-326.

Value

A list will be return, including mag, Rrup, lnY, sigma, specT, attenName, Vs30, period, iflag.

Examples

```
Gregor06(6, 20, 0, 760)
Gregor06(7, 10, 0, 760)
```

I14 *GMPE function for Idriss(2014)*

Description

I14 returns the ground-motion prediction with it sigma of Akkar et al.(2014) GMPE.

Usage

```
I14(Mag, Rrup, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Idriss, I. M. (2014). An NGA-West2 Empirical Model for Estimating the Horizontal Spectral Values Generated by Shallow Crustal Earthquakes. Earthquake Spectra, 30(3), 1155-1177. doi:10.1193/070613EQS195M Notes: Applicable Range (see Abstract): 5 <= M <= 8.5 Vs>=450 m/sec for Vs>1200 use Vs=1200 Rrup <= 150 km Mechanisms: Strike-slip and Norml (0) Reverse and Oblique (1)

Value

A list will be return, including mag, Rrup, ftype, Vs30, specT, period, lnY, sigma, iflag.

Examples

```
I14(6, 20, 0, 0, 760)
I14(7, 10, 0, 0, 760)
```

I14.tw.B01 *GMPE function for Idriss(2014)*

Description

I14.tw.B01 returns the ground-motion prediction with it sigma of Idriss, I. M. (2014) GMPE adjusted to Taiwan.

Usage

```
I14.tw.B01(Mag, Rrup, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).

Details

Idriss, I. M. (2014). An NGA-West2 Empirical Model for Estimating the Horizontal Spectral Values Generated by Shallow Crustal Earthquakes. Earthquake Spectra, 30(3), 1155-1177. doi:10.1193/070613EQS195M Notes: Applicable Range (see Abstract): 5 <= M <= 8.5 Vs>=450 m/sec for Vs>1200 use Vs=1200 Rrup <= 150 km Mechanisms: Strike-slip and Norml (0) Reverse and Oblique (1)

Value

A list will be return, including mag, Rrup, ftype, Vs30, specT, period, lnY, sigma, iflag.

Examples

```
I14.tw.B01(6, 20, 0, 0, 760)
I14.tw.B01(7, 10, 0, 0, 760)
```

I14.tw.C01 *GMPE function for Idriss(2014)*

Description

`I14.tw.C01` returns the ground-motion prediction with it sigma of Idriss, I. M. (2014) GMPE adjusted to Taiwan.

Usage

```
I14.tw.C01(Mag, Rrup, Prd = 0, ftype = 0, Vs30 = 760)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`ftype`	style of faulting.
`Vs30`	Vs30(m/s).

Details

Idriss, I. M. (2014). An NGA-West2 Empirical Model for Estimating the Horizontal Spectral Values Generated by Shallow Crustal Earthquakes. Earthquake Spectra, 30(3), 1155-1177. doi:10.1193/070613EQS195M Notes: Applicable Range (see Abstract): 5 <= M <= 8.5 Vs>=450 m/sec for Vs>1200 use Vs=1200 Rrup <= 150 km Mechanisms: Strike-slip and Norml (0) Reverse and Oblique (1)

Value

A list will be return, including mag, Rrup, ftype, Vs30, specT, period, lnY, sigma, iflag.

Examples

```
I14.tw.C01(6, 20, 0, 0, 760)
I14.tw.C01(7, 10, 0, 0, 760)
```

`I14.tw.E04` *GMPE function for for adjusted Idriss(2014) NGA-West2 model*

Description

`I14.tw.E04` returns the ground-motion prediction with it sigma of Idriss, I. M. (2014) GMPE adjusted to Taiwan.

Usage

```
I14.tw.E04(Mag, Rrup, Prd = 0, ftype = 0, Vs30 = 760, Ztor)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`ftype`	style of faulting.
`Vs30`	Vs30(m/s).
`Ztor`	Depth to the top of the finite rupture model (km).

Details

Idriss, I. M. (2014). An NGA-West2 Empirical Model for Estimating the Horizontal Spectral Values Generated by Shallow Crustal Earthquakes. Earthquake Spectra, 30(3), 1155-1177. doi:10.1193/070613EQS195M Notes: Applicable Range (see Abstract): 5 <= M <= 8.5 Vs>=450 m/sec for Vs>1200 use Vs=1200 Rrup <= 150 km Mechanisms: Strike-slip and Norml (0) Reverse and Oblique (1)

Value

A list will be return, including mag, Rrup, ftype, Vs30, specT, period, lnY, sigma, iflag.

Examples

```
I14.tw.E04(6, 20, 0, 0, 760, 0)
I14.tw.E04(7, 10, 0, 0, 760, 0)
```

KAAH15 *GMPE function for Kale et al.(2015)*

Description

KAAH15 returns the ground-motion prediction with it sigma of Kale et al.(2015) GMPE.

Usage

```
KAAH15(Mag, Rjb, Prd = 0, ftype = 0, Vs30 = 760, regionflag = 0)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	style of faulting.
Vs30	Vs30(m/s).
regionflag	0 for Turkey, 1 for Iran.

Details

Kale, O., S. Akkar, A. Ansari, and H. Hamzehloo (2015), A ground-Motion predictive model for Iran and Turkey for horizontal PGA, PGV, and 5 Possible Regional Effects, Bulletin of the Seismological Society of America, 105(2A), 963-980. doi:10.1785/0120140134

Value

A list will be return, including mag, Rjb, specT, period, lnY, sigma, iflag, ftype, Vs30, pgaref, region.

Examples

```
KAAH15(6, 20, 0, 0, 760, 0)
KAAH15(7, 10, 0, 0, 760, 0)
```

Kanno06 *GMPE function for Kanno et al.(2006)*

Description

Kanno06 returns the ground-motion prediction with it sigma of Kanno et al.(2006) GMPE.

Usage

```
Kanno06(Mag, Rrup, Prd, Vs30, depth)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
depth	hypocentral depth(km).

Details

Kanno, T., A. Narita, N. Morikawa, H. Fujiwara, and Y. Fukushima (2006), A new attenuation relation for strong ground motion in japan based on recorded data, Bulletin of the Seismological Society of America, 96(3), 879-897. doi:10.1785/0120050138

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag, Vs30, depth.

Examples

```
Kanno06(6, 20, 0, 760, 10)
Kanno06(7, 10, 0, 760, 10)
```

Lin09 *GMPE function for Lin (2009)*

Description

Lin09 returns the ground-motion prediction with it sigma of Lin (2009) GMPE.

Usage

```
Lin09(Mag, Rrup, Prd, Vs30 = 760, ftype)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s)
ftype	style of faulting.

Details

Lin, P.-S. (2009). Ground-motion attenuation relationship and path-effect study using Taiwan data set, Ph.D. Thesis, Nation Central University, Taiwan (in Chinese).

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, Vs30, iflag.

Examples

```
Lin09(6, 20, 0, 760, 0)
Lin09(7, 10, 0, 760, 0)
```

`Lin11` *GMPE function for Lin et al.(2011)*

Description

`Lin11` returns the ground-motion prediction (rock or soil site base on Vs30) with it sigma of Lin et al.(2011) GMPE.

Usage

```
Lin11(Mag, Rrup, Prd, hwflag = 0, Vs30 = 760)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`hwflag`	hanging-wall flag, 1 for hanging-wall.
`Vs30`	Vs30(m/s)

Details

Lin, P.-S., C.-T. Lee, C.-T. Cheng, and C.-H. Sung (2011), Response spectral attenuation relations for shallow crustal earthquakes in taiwan, Engineering Geology, 121(3-4), 150-164. doi:10.1016/j.enggeo.2011.04.019

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag.

Examples

```
Lin11(6, 20, 0, 0, 760)
Lin11(7, 10, 0, 0, 760)
```

Lin11Rock *GMPE function for Lin et al.(2011)*

Description

Lin11Rock returns the ground-motion prediction (rock site) with it sigma of Lin et al.(2011) GMPE.

Usage

```
Lin11Rock(Mag, Rrup, Prd, hwflag = 0)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
hwflag	hanging-wall flag, 1 for hanging-wall.

Details

Lin, P.-S., C.-T. Lee, C.-T. Cheng, and C.-H. Sung (2011), Response spectral attenuation relations for shallow crustal earthquakes in taiwan, Engineering Geology, 121(3-4), 150-164. doi:10.1016/j.enggeo.2011.04.019

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag.

Examples

```
Lin11Rock(6, 20, 0, 0)
Lin11Rock(7, 10, 0, 0)
```

Lin11Soil *GMPE function for Lin et al.(2011)*

Description

Lin11Soil returns the ground-motion prediction (soil site) with it sigma of Lin et al.(2011) GMPE.

Usage

```
Lin11Soil(Mag, Rrup, Prd, hwflag = 0)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
hwflag	hanging-wall flag, 1 for hanging-wall.

Details

Lin, P.-S., C.-T. Lee, C.-T. Cheng, and C.-H. Sung (2011), Response spectral attenuation relations for shallow crustal earthquakes in taiwan, Engineering Geology, 121(3-4), 150-164. doi:10.1016/j.enggeo.2011.04.019

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag.

Examples

```
Lin11Soil(6, 20, 0, 0)
Lin11Soil(7, 10, 0, 0)
```

`LL08` *GMPE function for Lin and Lee(2008)*

Description

`LL08` returns the ground-motion prediction (Rock or Soil site base on Vs30) with it sigma of Lin and Lee(2008) GMPE.

Usage

```
LL08(Mag, Rrup, depth, ftype, Prd, Vs30 = 760)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`depth`	hypocentral depth(km).
`ftype`	fytpe=0 for interface, ftype=1 for intraslab
`Prd`	Period of spectral acceleration.
`Vs30`	Vs30(m/s)

Details

Lin, P., and C. Lee (2008), Ground-motion attenuation relationships for subduction-zone earthquakes in northeastern taiwan, Bulletin of the Seismological Society of America, 98(1), 220-240. doi:10.1785/0120060002

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag, depth, ftype.

Examples

```
LL08(6, 20, 10, 0, 0, 760)
LL08(7, 10, 10, 0, 0, 760)
```

LL08.C02 *GMPE function for adjusted Lin and Lee(2008)*

Description

LL08.C02 returns the ground-motion prediction with it sigma of adjusted Lin and Lee(2008) GMPE.

Usage

```
LL08.C02(Mag, Rrup, Ztor, ftype = 0, Prd, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	fytpe=0 for interface, ftype=1 for intraslab
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s)

Details

Lin, P., and C. Lee (2008), Ground-motion attenuation relationships for subduction-zone earthquakes in northeastern taiwan, Bulletin of the Seismological Society of America, 98(1), 220-240. doi:10.1785/0120060002

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag, Ztor, ftype, Vs30.

Examples

```
LL08.C02(6, 20, 10, 0, 0, 760)
LL08.C02(7, 10, 10, 0, 0, 760)
```

LL08.F04 *GMPE function for adjusted Lin and Lee(2008)*

Description

LL08.F04 returns the ground-motion prediction with it sigma of adjusted Lin and Lee(2008) GMPE.

Usage

```
LL08.F04(Mag, Rrup, Ztor, ftype = 0, Prd, Vs30 = 760)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
ftype	fytpe=0 for interface, ftype=1 for intraslab
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s)

Details

Lin, P., and C. Lee (2008), Ground-motion attenuation relationships for subduction-zone earthquakes in northeastern taiwan, Bulletin of the Seismological Society of America, 98(1), 220-240. doi:10.1785/0120060002

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag, Ztor, ftype, Vs30.

Examples

```
LL08.F04(6, 20, 10, 0, 0, 760)
LL08.F04(7, 10, 10, 0, 0, 760)
```

LL08Rock *GMPE function for Lin and Lee(2008)*

Description

LL08Rock returns the ground-motion prediction (rock site) with it sigma of Lin and Lee(2008) GMPE.

Usage

```
LL08Rock(Mag, Rrup, depth, ftype, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
depth	hypocentral depth(km).
ftype	fytpe=0 for interface, ftype=1 for intraslab
Prd	Period of spectral acceleration.

Details

Lin, P., and C. Lee (2008), Ground-motion attenuation relationships for subduction-zone earthquakes in northeastern taiwan, Bulletin of the Seismological Society of America, 98(1), 220-240. doi:10.1785/0120060002

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag, depth, ftype.

Examples

```
LL08Rock(6, 20, 10, 0, 0)
LL08Rock(7, 10, 10, 0, 0)
```

`LL08Soil` *GMPE function for Lin and Lee(2008)*

Description

`LL08Soil` returns the ground-motion prediction (soil site) with it sigma of Lin and Lee(2008) GMPE.

Usage

```
LL08Soil(Mag, Rrup, depth, ftype, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`depth`	hypocentral depth(km).
`ftype`	fytpe=0 for interface, ftype=1 for intraslab
`Prd`	Period of spectral acceleration.

Details

Lin, P., and C. Lee (2008), Ground-motion attenuation relationships for subduction-zone earthquakes in northeastern taiwan, Bulletin of the Seismological Society of America, 98(1), 220-240. doi:10.1785/0120060002

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, iflag, depth, ftype.

Examples

```
LL08Soil(6, 20, 10, 0, 0)
LL08Soil(7, 10, 10, 0, 0)
```

Loh96 *GMPE function for Loh et al.(1996) PGA only*

Description

Loh96 returns the ground-motion prediction (rock site) with it sigma of Loh et al.(1996) GMPE.

Usage

```
Loh96(Mag, Rrup)
```

Arguments

Mag	Earthquake local magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.

Value

A list will be return, including mag, Rrup, lnY, sigma, period.

Examples

```
Loh96(6, 20)
Loh96(7, 10)
```

MBR17 *GMPE function for Montalva et al.(2017)*

Description

MBR17 returns the ground-motion prediction with it sigma of Montalva et al.(2017) GMPE.

Usage

```
MBR17(Mag, Rrup, Prd, ftype = 0, Vs30, forearc = 1, depth, Rhypo)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface (use rupture distance), ftype=1 for intraslab (use hypocentral distance).
Vs30	Vs30(m/s).
forearc	0 = Forearc site, 1 = Backarc site.
depth	hypocentral depth(km).
Rhypo	hypocentral distance(km).

Details

Montalva, G. A., Bastias, N., and Rodriguez-Marek, A. (2017). Ground-Motion Prediction Equation for the Chilean Subduction Zone. Bulletin of the Seismological Society of America, 107(2), 901-911. doi:10.1785/0120160221

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo.

Examples

```
MBR17(6, 20, 0, 0, 760, 1, 10, 20)
MBR17(7, 10, 0, 0, 760, 1, 10, 20)
```

Mcverry2006Sub *GMPE function for Mcverry et al.(2006) Subduction GMPE.*

Description

`Mcverry2006Sub` returns the ground-motion prediction with it sigma of Mcverry et al.(2006) Subduction GMPE.

Usage

```
Mcverry2006Sub(
  Mag,
  Rrup,
  Prd,
  ftype = 0,
  Ztor,
  Dip,
  rupwidth,
  depth,
  Sc = 0,
  Sd = 0
)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`ftype`	fytpe=0 for interface, ftype=1 for intraslab.
`Ztor`	Depth to the top of the finite rupture model (km).
`Dip`	Dip angle of the fault plane.
`rupwidth`	Down-dip rupture width (km).
`depth`	hypocentral depth(km).
`Sc`	1 for site class C, 0 for others.
`Sd`	1 for site class D, 0 for others.

Details

Mcverry, G. H., Zhao, J. X., Abrahamson, N. A., and Somerville, P. G. (2006). New Zealand acceleration response spectrum attenuation relations for crustal and subduction zone earthquakes, Bulletin of the New Zealand Society for Earthquake Engineering, 39(1), 1-58.

Value

A list will be return, including mag, Rrup, Rjb, ftype, specT, period, lnY, sigma, iflag, Vs30, Ztor, Z25, dip, depth, hwflag, Rx, rupwidth, regionflag, phi, tau.

Examples

```
Mcverry2006Sub(6, 20, 0, 0, 0, 90, 10, 10, 0, 0)
Mcverry2006Sub(7, 20, 0, 0, 0, 90, 10, 10, 0, 0)
```

NCREE2011 *GMPE function for NCREE(2011)*

Description

NCREE2011 returns the ground-motion prediction (rock site, Vs30>=360m/s) with it sigma of NCREE(2011) GMPE.

Usage

```
NCREE2011(Mag, Rrup, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.

Details

National Center for Research on Earthquake Engineering (NCREE) (2011). Research on area source and attenuation law parameters and nuclear power plant site response measurement. Commissioned research report by the Atomic Energy Council, Institute of Nuclear Energy Research, Taiwan.

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma.

Examples

```
NCREE2011(6, 20, 0)
NCREE2011(7, 10, 0)
```

periods.T *Periods of NGA-West GMPE*

Description

Periods of NGA-West GMPE

Usage

```
periods.T
```

Format

A vector with 105 values.

PhungCru17 *GMPE function for Phung et al.(2017)*

Description

PhungCru17 returns the ground-motion prediction with it sigma of Phung et al.(2017) GMPE.

Usage

```
PhungCru17(
  Mag,
  Rrup,
  Prd,
  Vs30,
  Dip,
  Ztor,
  ftype = 0,
  Z1.0,
  Vs30_class = 1,
  msasflag = 0,
  regionflag = 1
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Dip	Dip angle of the fault plane.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.

Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Vs30_class	1 measured, 0 estimated.
msasflag	Mainshock and Aftershocks flag, 0 = Mainshocks, 1 = Aftershocks.
regionflag	0 Global, 1 Taiwan

Details

Phung et al.(2017) Crustal GMPE

Value

A list will be return, including m, Rrup, specT, period2, lnY, sigma, iflag, vs, Delta, DTor, Ftype, depthvs10, vs30_class, regionflag, msasflag, phi, tau.

Examples

```
PhungCru17(6, 20, 0, 760, 90, 0, 0, 0.5, 1, 0,  1)
PhungCru17(7, 20, 0, 760, 90, 0, 0, 0.5, 1, 0,  1)
```

PhungCru18 *GMPE function for Phung et al.(2018)*

Description

PhungCru18 returns the ground-motion prediction with it sigma of Phung et al.(2018) GMPE.

Usage

```
PhungCru18(
  Mag,
  Rrup,
  Rjb,
  Rx,
  Prd,
  Vs30,
  Dip,
  Ztor,
  ftype = 0,
  Z1.0,
  Vs30_class = 1,
  regionflag = 1,
  hwflag = 0
)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Rjb	Joyner and Boore distance(km), Numeric.
Rx	Horizontal distance(km) from top edge of rupture. Measured perpendicular to the fault strike.
Prd	Period of spectral acceleration.
Vs30	Vs30(m/s).
Dip	Dip angle of the fault plane.
Ztor	Depth to the top of the finite rupture model (km).
ftype	style of faulting.
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Vs30_class	1 measured, 0 estimated.
regionflag	0 Global, 1 Taiwan
hwflag	hanging-wall flag, 1 for hanging-wall.

Details

Phung et al.(2018) Crustal GMPE

Value

A list will be return, including m, Rrup, Rjb, Rx, specT, period2, lnY, sigma, iflag, vs, Delta, DTor, Ftype, depthvs10, vs30_class, regionflag, phi, tau, HWflag, Rx.

Examples

```
PhungCru18(6, 20, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 1, 0)
PhungCru18(7, 20, 20, 20, 0, 760, 90, 0, 0, 0.5, 1, 1, 0)
```

PhungSub17 *GMPE function for Phung et al.(2017)*

Description

PhungSub17 returns the ground-motion prediction with it sigma of Phung et al.(2017) GMPE.

Usage

```
PhungSub17(Mag, Rrup, Prd, ftype = 0, Vs30, Z1.0, Ztor, regionflag = 1)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface, ftype=1 for intraslab.
Vs30	Vs30(m/s).
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Ztor	Depth to the top of the finite rupture model (km).
regionflag	0 Japan, 1 Taiwan

Details

Phung et al.(2017) Subduction GMPE

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo.

Examples

```
PhungSub17(6, 20, 0, 0, 760, 0.5, 0, 1)
PhungSub17(7, 10, 0, 1, 760, 0.5, 0, 1)
```

PhungSub18 *GMPE function for Phung et al.(2018)*

Description

PhungSub18 returns the ground-motion prediction with it sigma of Phung et al.(2018) GMPE.

Usage

```
PhungSub18(Mag, Rrup, Prd, ftype = 0, Vs30, Z1.0, Ztor, regionflag = 1)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface, ftype=1 for intraslab.
Vs30	Vs30(m/s).
Z1.0	the depth to the shear-wave velocity horizon of 1.0(km/s).
Ztor	Depth to the top of the finite rupture model (km).
regionflag	0 Japan, 1 Taiwan

Details

Phung et al.(2018) Subduction GMPE

Value

A list will be return, including mag, ftype, Rrup, Vs30, lnY, sigma, specT, period, iflag, forearc, depth, Rhypo.

Examples

```
PhungSub18(6, 20, 0, 0, 760, 0.5, 0, 1)
PhungSub18(7, 10, 0, 1, 760, 0.5, 0, 1)
```

Sinotech12 *GMPE function for Sinotech(2012)*

Description

`Sinotech12` returns the ground-motion prediction with it sigma of Sinotech(2012) GMPE.

Usage

```
Sinotech12(Mag, Rrup, Prd, Vs30 = 760, ftype)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Prd`	Period of spectral acceleration.
`Vs30`	Vs30(m/s)
`ftype`	style of faulting.

Details

Sinotech (2012). (in Chinese).

Value

A list will be return, including mag, Rrup, specT, period, lnY, sigma, Vs30, iflag.

Examples

```
Sinotech12(6, 20, 0, 760, 0)
Sinotech12(7, 10, 0, 760, 0)
```

Sub.Inter.Com.001 *GMPE function for Subduction interface common form 001 (2017)*

Description

`Sub.Inter.Com.001` returns the ground-motion prediction with it sigma of Subduction interface Common form 001 GMPE.

Usage

```
Sub.Inter.Com.001(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 001

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.001(6, 20, 5, 0)
Sub.Inter.Com.001(7, 20, 2, 0)
```

Sub.Inter.Com.002 *GMPE function for Subduction interface common form 002 (2017)*

Description

`Sub.Inter.Com.002` returns the ground-motion prediction with it sigma of Subduction interface Common form 002 GMPE.

Usage

```
Sub.Inter.Com.002(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 002

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.002(6, 20, 5, 0)
Sub.Inter.Com.002(7, 20, 2, 0)
```

Sub.Inter.Com.003 *GMPE function for Subduction interface common form 003 (2017)*

Description

`Sub.Inter.Com.003` returns the ground-motion prediction with it sigma of Subduction interface Common form 003 GMPE.

Usage

```
Sub.Inter.Com.003(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 003

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.003(6, 20, 5, 0)
Sub.Inter.Com.003(7, 20, 2, 0)
```

Sub.Inter.Com.004 *GMPE function for Subduction interface common form 004 (2017)*

Description

`Sub.Inter.Com.004` returns the ground-motion prediction with it sigma of Subduction interface Common form 004 GMPE.

Usage

```
Sub.Inter.Com.004(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 004

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.004(6, 20, 5, 0)
Sub.Inter.Com.004(7, 20, 2, 0)
```

Sub.Inter.Com.005 *GMPE function for Subduction interface common form 005 (2017)*

Description

`Sub.Inter.Com.005` returns the ground-motion prediction with it sigma of Subduction interface Common form 005 GMPE.

Usage

```
Sub.Inter.Com.005(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 005

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.005(6, 20, 5, 0)
Sub.Inter.Com.005(7, 20, 2, 0)
```

Sub.Inter.Com.006 *GMPE function for Subduction interface common form 006 (2017)*

Description

`Sub.Inter.Com.006` returns the ground-motion prediction with it sigma of Subduction interface Common form 006 GMPE.

Usage

```
Sub.Inter.Com.006(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 006

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.006(6, 20, 5, 0)
Sub.Inter.Com.006(7, 20, 2, 0)
```

Sub.Inter.Com.007 *GMPE function for Subduction interface common form 007 (2017)*

Description

`Sub.Inter.Com.007` returns the ground-motion prediction with it sigma of Subduction interface Common form 007 GMPE.

Usage

```
Sub.Inter.Com.007(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 007

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.007(6, 20, 5, 0)
Sub.Inter.Com.007(7, 20, 2, 0)
```

Sub.Inter.Com.008 *GMPE function for Subduction interface common form 008 (2017)*

Description

`Sub.Inter.Com.008` returns the ground-motion prediction with it sigma of Subduction interface Common form 008 GMPE.

Usage

```
Sub.Inter.Com.008(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 008

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.008(6, 20, 5, 0)
Sub.Inter.Com.008(7, 20, 2, 0)
```

`Sub.Inter.Com.009` *GMPE function for Subduction interface common form 009 (2017)*

Description

`Sub.Inter.Com.009` returns the ground-motion prediction with it sigma of Subduction interface Common form 009 GMPE.

Usage

```
Sub.Inter.Com.009(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 009

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.009(6, 20, 5, 0)
Sub.Inter.Com.009(7, 20, 2, 0)
```

Sub.Inter.Com.010 *GMPE function for Subduction interface common form 010 (2017)*

Description

Sub.Inter.Com.010 returns the ground-motion prediction with it sigma of Subduction interface Common form 010 GMPE.

Usage

```
Sub.Inter.Com.010(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction interface common form 010

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.010(6, 20, 5, 0)
Sub.Inter.Com.010(7, 20, 2, 0)
```

Sub.Inter.Com.011 *GMPE function for Subduction interface common form 011 (2017)*

Description

Sub.Inter.Com.011 returns the ground-motion prediction with it sigma of Subduction interface Common form 011 GMPE.

Usage

```
Sub.Inter.Com.011(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction interface common form 011

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.011(6, 20, 5, 0)
Sub.Inter.Com.011(7, 20, 2, 0)
```

Sub.Inter.Com.012 *GMPE function for Subduction interface common form 012 (2017)*

Description

`Sub.Inter.Com.012` returns the ground-motion prediction with it sigma of Subduction interface Common form 012 GMPE.

Usage

```
Sub.Inter.Com.012(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 012

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.012(6, 20, 5, 0)
Sub.Inter.Com.012(7, 20, 2, 0)
```

Sub.Inter.Com.013 *GMPE function for Subduction interface common form 013 (2017)*

Description

Sub.Inter.Com.013 returns the ground-motion prediction with it sigma of Subduction interface Common form 013 GMPE.

Usage

```
Sub.Inter.Com.013(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction interface common form 013

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.013(6, 20, 5, 0)
Sub.Inter.Com.013(7, 20, 2, 0)
```

Sub.Inter.Com.014 *GMPE function for Subduction interface common form 014 (2017)*

Description

Sub.Inter.Com.014 returns the ground-motion prediction with it sigma of Subduction interface Common form 014 GMPE.

Usage

```
Sub.Inter.Com.014(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction interface common form 014

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.014(6, 20, 5, 0)
Sub.Inter.Com.014(7, 20, 2, 0)
```

Sub.Inter.Com.015 *GMPE function for Subduction interface common form 015 (2017)*

Description

`Sub.Inter.Com.015` returns the ground-motion prediction with it sigma of Subduction interface Common form 015 GMPE.

Usage

```
Sub.Inter.Com.015(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction interface common form 015

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.015(6, 20, 5, 0)
Sub.Inter.Com.015(7, 20, 2, 0)
```

Sub.Inter.Com.016 *GMPE function for Subduction interface common form 016 (2017)*

Description

Sub.Inter.Com.016 returns the ground-motion prediction with it sigma of Subduction interface Common form 016 GMPE.

Usage

```
Sub.Inter.Com.016(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction interface common form 016

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.016(6, 20, 5, 0)
Sub.Inter.Com.016(7, 20, 2, 0)
```

Sub.Inter.Com.017 *GMPE function for Subduction interface common form 017 (2017)*

Description

Sub.Inter.Com.017 returns the ground-motion prediction with it sigma of Subduction interface Common form 017 GMPE.

Usage

```
Sub.Inter.Com.017(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction interface common form 017

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Inter.Com.017(6, 20, 5, 0)
Sub.Inter.Com.017(7, 20, 2, 0)
```

Sub.Intra.Com.001 *GMPE function for Subduction intraslab common form 001 (2017)*

Description

`Sub.Intra.Com.001` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 001 GMPE.

Usage

```
Sub.Intra.Com.001(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 001

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.001(6, 20, 5, 0)
Sub.Intra.Com.001(7, 20, 2, 0)
```

Sub.Intra.Com.002 *GMPE function for Subduction intraslab common form 002 (2017)*

Description

`Sub.Intra.Com.002` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 002 GMPE.

Usage

```
Sub.Intra.Com.002(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 002

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.002(6, 20, 5, 0)
Sub.Intra.Com.002(7, 20, 2, 0)
```

Sub.Intra.Com.003 *GMPE function for Subduction intraslab common form 003 (2017)*

Description

`Sub.Intra.Com.003` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 003 GMPE.

Usage

```
Sub.Intra.Com.003(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 003

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.003(6, 20, 5, 0)
Sub.Intra.Com.003(7, 20, 2, 0)
```

Sub.Intra.Com.004 *GMPE function for Subduction intraslab common form 004 (2017)*

Description

`Sub.Intra.Com.004` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 004 GMPE.

Usage

```
Sub.Intra.Com.004(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 004

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.004(6, 20, 5, 0)
Sub.Intra.Com.004(7, 20, 2, 0)
```

Sub.Intra.Com.005 *GMPE function for Subduction intraslab common form 005 (2017)*

Description

Sub.Intra.Com.005 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 005 GMPE.

Usage

```
Sub.Intra.Com.005(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 005

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.005(6, 20, 5, 0)
Sub.Intra.Com.005(7, 20, 2, 0)
```

Sub.Intra.Com.006 *GMPE function for Subduction intraslab common form 006 (2017)*

Description

Sub.Intra.Com.006 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 006 GMPE.

Usage

```
Sub.Intra.Com.006(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 006

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.006(6, 20, 5, 0)
Sub.Intra.Com.006(7, 20, 2, 0)
```

`Sub.Intra.Com.007` *GMPE function for Subduction intraslab common form 007 (2017)*

Description

`Sub.Intra.Com.007` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 007 GMPE.

Usage

```
Sub.Intra.Com.007(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 007

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.007(6, 20, 5, 0)
Sub.Intra.Com.007(7, 20, 2, 0)
```

Sub.Intra.Com.008 *GMPE function for Subduction intraslab common form 008 (2017)*

Description

Sub.Intra.Com.008 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 008 GMPE.

Usage

```
Sub.Intra.Com.008(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 008

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.008(6, 20, 5, 0)
Sub.Intra.Com.008(7, 20, 2, 0)
```

Sub.Intra.Com.009 *GMPE function for Subduction intraslab common form 009 (2017)*

Description

Sub.Intra.Com.009 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 009 GMPE.

Usage

```
Sub.Intra.Com.009(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 009

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.009(6, 20, 5, 0)
Sub.Intra.Com.009(7, 20, 2, 0)
```

Sub.Intra.Com.010 *GMPE function for Subduction intraslab common form 010 (2017)*

Description

`Sub.Intra.Com.010` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 010 GMPE.

Usage

```
Sub.Intra.Com.010(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 010

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.010(6, 20, 5, 0)
Sub.Intra.Com.010(7, 20, 2, 0)
```

Sub.Intra.Com.011 *GMPE function for Subduction intraslab common form 011 (2017)*

Description

Sub.Intra.Com.011 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 011 GMPE.

Usage

```
Sub.Intra.Com.011(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 011

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.011(6, 20, 5, 0)
Sub.Intra.Com.011(7, 20, 2, 0)
```

Sub.Intra.Com.012 *GMPE function for Subduction intraslab common form 012 (2017)*

Description

Sub.Intra.Com.012 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 012 GMPE.

Usage

```
Sub.Intra.Com.012(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 012

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.012(6, 20, 5, 0)
Sub.Intra.Com.012(7, 20, 2, 0)
```

Sub.Intra.Com.013 *GMPE function for Subduction intraslab common form 013 (2017)*

Description

`Sub.Intra.Com.013` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 013 GMPE.

Usage

```
Sub.Intra.Com.013(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 013

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.013(6, 20, 5, 0)
Sub.Intra.Com.013(7, 20, 2, 0)
```

Sub.Intra.Com.014 *GMPE function for Subduction intraslab common form 014 (2017)*

Description

Sub.Intra.Com.014 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 014 GMPE.

Usage

```
Sub.Intra.Com.014(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 014

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.014(6, 20, 5, 0)
Sub.Intra.Com.014(7, 20, 2, 0)
```

Sub.Intra.Com.015 *GMPE function for Subduction intraslab common form 015 (2017)*

Description

Sub.Intra.Com.015 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 015 GMPE.

Usage

```
Sub.Intra.Com.015(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 015

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.015(6, 20, 5, 0)
Sub.Intra.Com.015(7, 20, 2, 0)
```

Sub.Intra.Com.016 *GMPE function for Subduction intraslab common form 016 (2017)*

Description

`Sub.Intra.Com.016` returns the ground-motion prediction with it sigma of Subduction intraslab Common form 016 GMPE.

Usage

```
Sub.Intra.Com.016(Mag, Rrup, Ztor, Prd)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`Ztor`	Depth to the top of the finite rupture model (km).
`Prd`	Period of spectral acceleration.

Details

Subduction intraslab common form 016

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.016(6, 20, 5, 0)
Sub.Intra.Com.016(7, 20, 2, 0)
```

Sub.Intra.Com.017 *GMPE function for Subduction intraslab common form 017 (2017)*

Description

Sub.Intra.Com.017 returns the ground-motion prediction with it sigma of Subduction intraslab Common form 017 GMPE.

Usage

```
Sub.Intra.Com.017(Mag, Rrup, Ztor, Prd)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Ztor	Depth to the top of the finite rupture model (km).
Prd	Period of spectral acceleration.

Details

Subduction intraslab common form 017

Value

A list will be return, including mag, Rrup, Ztor, pecT, lnY, sigma, iflag.

Examples

```
Sub.Intra.Com.017(6, 20, 5, 0)
Sub.Intra.Com.017(7, 20, 2, 0)
```

Youngs97Rock *GMPE function for Youngs et al.(1997)*

Description

Youngs97Rock returns the ground-motion prediction (rock site) with it sigma of Youngs et al.(1997) GMPE.

Usage

```
Youngs97Rock(Mag, Rrup, Prd, ftype = 0, depth)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface, ftype=1 for intraslab
depth	hypocentral depth(km).

Details

Youngs, R. R., S.-J. Chiou, W. J. Silva, and J. R. Humphrey (1997), Strong ground motion attenuation relationships for subduction zone earthquakes, Seismological Research Letters, 68(1), 58-73. doi:10.1785/gssrl.68.1.58

Value

A list will be return, including mag, Rrup, lnY, sigma, period, specT, ftype, depth, iflag.

Examples

```
Youngs97Rock(6, 20, 0, 0, 10)
Youngs97Rock(7, 10, 0, 0, 10)
```

Youngs97Soil *GMPE function for Youngs et al.(1997)*

Description

`Youngs97Soil` returns the ground-motion prediction (soil site) with it sigma of Youngs et al.(1997) GMPE.

Usage

```
Youngs97Soil(Mag, Rrup, Prd, ftype = 0, depth)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
Prd	Period of spectral acceleration.
ftype	fytpe=0 for interface, ftype=1 for intraslab
depth	hypocentral depth(km).

Details

Youngs, R. R., S.-J. Chiou, W. J. Silva, and J. R. Humphrey (1997), Strong ground motion attenuation relationships for subduction zone earthquakes, Seismological Research Letters, 68(1), 58-73. doi:10.1785/gssrl.68.1.58

Value

A list will be return, including mag, Rrup, lnY, sigma, period, specT, ftype, depth, iflag.

Examples

```
Youngs97Soil(6, 20, 0, 0, 10)
Youngs97Soil(7, 10, 0, 0, 10)
```

`Zh06` *GMPE function for Zhao et al.(2006)*

Description

`Zh06` returns the ground-motion prediction with it sigma of Zhao et al.(2006) GMPE.

Usage

```
Zh06(Mag, Rrup, ftype = 0, sclass = 1, Prd = 0, sourceclass = 0, depth = 10)
```

Arguments

`Mag`	Earthquake momnet magnitude, Numeric.
`Rrup`	Rupture distance(km), Numeric.
`ftype`	style of faulting.
`sclass`	site condition.
`Prd`	Period of spectral acceleration.
`sourceclass`	source type, 0 for crustal, 1 for subduction interface, 2 for subduction intraslab.
`depth`	focal depth(km).

Details

Zhao, J. et al. (2006), Attenuation relations of strong ground motion in japan using site classification based on predominant period, Bulletin of the Seismological Society of America, 96(3), 898-913. doi:10.1785/0120050122

Value

A list will be return, including mag, Rrup, ftype, lnY, sigma, sclass, specT, attenName, period, iflag, sourcetype, depth, phi, tau.

Examples

```
Zh06(6, 20, 0, 1.0, 0, 0.0, 10)
Zh06(7, 10, 0, 1.0, 0, 0.0, 10)
```

Zh16Cru *GMPE function for Zhao et al.(2016)*

Description

Zh16Cru returns the ground-motion prediction with it sigma of Zhao et al.(2016) GMPE.

Usage

```
Zh16Cru(Mag, Rrup, ftype = 0, sclass = 1, Prd = 0, sourceclass = 0, depth = 10)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
ftype	style of faulting, -1 for Normal, -0.5 for Normal-Oblique, 0 for Strike-Slip, 0.5 for Reverse-Oblique and 1 for Reverse.
sclass	site condition, 0 for Hard Rock, 1 for SC I, 2 for SC II, 3 for SC III, 4 for SC IV.
Prd	Period of spectral acceleration.
sourceclass	source type, 0 for crustal, 3 for Upper mantle.
depth	focal depth(km).

Details

Zhao, J. et al. (2016), Ground-Motion Prediction Equations for Shallow Crustal and Upper-Mantle Earthquakes in Japan Using Site Class and Simple Geometric Attenuation Functions, Bulletin of the Seismological Society of America, 106(4), 1552-1569. doi:10.1785/0120150063

Value

A list will be return, including mag, Rrup, ftype, lnY, sigma, sclass, specT, attenName, period, iflag, sourcetype, depth, phi, tau.

Examples

```
Zh16Cru(6, 20, 0, 1.0, 0, 0.0, 10)
Zh16Cru(7, 10, 0, 1.0, 0, 0.0, 10)
```

Zh16Inter *GMPE function for Zhao et al.(2016)*

Description

Zh06Inter returns the ground-motion prediction with it sigma of Zhao et al.(2016) GMPE.

Usage

```
Zh16Inter(Mag, Rrup, sclass = 1, Prd = 0, depth = 10)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
sclass	site condition, 0 for Hard Rock, 1 for SC I, 2 for SC II, 3 for SC III, 4 for SC IV.
Prd	Period of spectral acceleration.
depth	focal depth(km).

Details

Zhao, J. et al. (2016),Ground-Motion Prediction Equations for Subduction Interface Earthquakes in Japan Using Site Class and Simple Geometric Attenuation Functions, Bulletin of the Seismological Society of America, 106(4), 1518-1534. doi:10.1785/0120150034

Value

A list will be return, including mag, Rrup, ftype, lnY, sigma, sclass, specT, attenName, period, iflag, sourcetype, depth, phi, tau.

Examples

```
Zh16Inter(6, 20, 1.0, 0, 10)
Zh16Inter(7, 10, 1.0, 0, 10)
```

Zh16Slab *GMPE function for Zhao et al.(2016)*

Description

Zh06Slab returns the ground-motion prediction with it sigma of Zhao et al.(2016) GMPE.

Usage

```
Zh16Slab(Mag, Rrup, sclass = 1, Prd = 0, depth = 10)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
sclass	site condition, 0 for Hard Rock, 1 for SC I, 2 for SC II, 3 for SC III, 4 for SC IV.
Prd	Period of spectral acceleration.
depth	focal depth(km).

Details

Zhao, J. et al. (2016),Ground-Motion Prediction Equations for Subduction Slab Earthquakes in Japan Using Site Class and Simple Geometric Attenuation Functions, Bulletin of the Seismological Society of America, 106(4), 1535-1551. doi:10.1785/0120150056

Value

A list will be return, including mag, Rrup, ftype, lnY, sigma, sclass, specT, attenName, period, iflag, sourcetype, depth, phi, tau.

Examples

```
Zh16Slab(6, 20, 1.0, 0, 10)
Zh16Slab(7, 10, 1.0, 0, 10)
```

Zh16Sub *GMPE function for Zhao et al.(2016) Subduction GMPE*

Description

Zh16Sub returns the ground-motion prediction with it sigma of Zhao et al.(2016) GMPE.

Usage

```
Zh16Sub(Mag, Rrup, sclass = 1, Prd = 0, sourceclass = 1, depth = 10)
```

Arguments

Mag	Earthquake momnet magnitude, Numeric.
Rrup	Rupture distance(km), Numeric.
sclass	site condition, 0 for Hard Rock, 1 for SC I, 2 for SC II, 3 for SC III, 4 for SC IV.
Prd	Period of spectral acceleration.
sourceclass	source type, 1 for Subduction - Interface, 2 for Subduction - Slab.
depth	focal depth(km).

Details

Zhao, J. et al. (2016),Ground-Motion Prediction Equations for Subduction Interface Earthquakes in Japan Using Site Class and Simple Geometric Attenuation Functions, Bulletin of the Seismological Society of America, 106(4), 1518-1534. doi:10.1785/0120150034 #'Zhao, J. et al. (2016),Ground-Motion Prediction Equations for Subduction Slab Earthquakes in Japan Using Site Class and Simple Geometric Attenuation Functions, Bulletin of the Seismological Society of America, 106(4), 1535-1551. doi:10.1785/0120150056

Value

A list will be return, including mag, Rrup, ftype, lnY, sigma, sclass, specT, attenName, period, iflag, sourcetype, depth, phi, tau.

Examples

```
Zh16Sub(6, 20, 1.0, 0, 1.0, 10)
Zh16Sub(7, 10, 1.0, 0, 1.0, 10)
```